KU LEUVEN

FACULTY OF ECONOMICS AND BUSINESS

Shifting drivers of stock returns: machine learning evidence on time-varying predictability

How do top firm-level characteristics identified by machine learning techniques as predictors of stock returns vary over time?

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Shifting drivers of stock returns: machine learning evidence on time-varying predictability

How do top firm level-characteristics identified by machine learning techniques as predictors of stock returns vary over time?

Whether the firm characteristics that predict cross-sectional stock returns are stable over time or shift across market regimes is an open empirical question. Classical predictors such as value and momentum, once central to factor investing, have weakened substantially in recent decades, while size and liquidity-based signals appear more persistent. This thesis examines how the importance of firm-level characteristics as predictors of stock returns varies over time. Machine learning is well-suited to tracking this time variation because it can simultaneously evaluate a large set of predictors, capture nonlinear interactions, and apply disciplined out-of-sample validation, providing a comprehensive view across many features rather than a small number of pre-selected factors.

Three machine learning models, ElasticNet, Random Forest, and a feed-forward Neural Network, are estimated on a panel of US stocks from 1970 to 2024 using rolling 10-year training windows with per-decade hyperparameter tuning. The rolling design generates separate feature importance rankings for each prediction year, allowing predictor importance to be tracked across decades rather than averaged over the full sample.

The results suggest that predictor importance is regime-dependent rather than stable. Size-related characteristics persist as top predictors in every decade, but the secondary predictors shift substantially over time: momentum dominates the secondary tier in the 1980s, liquidity rises through the 2000s, and accounting fundamentals, including a partial recovery of the book-to-market ratio, gain prominence in the 2010s and 2020s. These findings complement the full-sample rankings of Gu et al. (2020) by indicating that feature importance may evolve meaningfully when predictability is examined dynamically rather than averaged across decades. An alpha-grid analysis suggests these time-varying shifts are not driven solely by model choice.

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1. General introduction

For decades, value investing was widely regarded as one of the most reliable strategies for generating cross-sectional excess returns. The seminal work of Fama and French (1992, 1993) established a robust positive relationship between book-to-market ratios and average stock returns, and the strategy of buying high book-to-market firms and shorting low book-to-market firms became a cornerstone of academic asset pricing and quantitative investment practice. Yet over the past two decades, this once-dominant predictor has weakened substantially. The book-to-market premium has compressed; high-profile quantitative funds have closed underperforming value strategies; and a growing literature debates whether the value effect has merely paused, structurally declined, or shifted into other forms (Fama & French, 2021; Israel et al., 2021; Gonçalves & Leonard, 2023). The story of value investing is, in essence, a story about time variation: a predictor that worked spectacularly in one era no longer behaves the same way in another.

This pattern is not unique to value. Momentum, the second pillar of factor investing introduced by Jegadeesh and Titman (1993), has also weakened in recent decades, with returns deteriorating after publication and arbitrage capital flowing into momentum strategies (McLean & Pontiff, 2016). Hou et al. (2020) document that a substantial fraction of published return anomalies fail to replicate under stricter statistical and portfolio-construction standards, and Linnainmaa and Roberts (2018) show that many accounting-based predictors weaken sharply out of sample. Meanwhile, other characteristics, such as illiquidity (Amihud, 2002), or intangible-investment intensity (Dyer et al., 2025), have grown in relative importance as market structure and the information environment have evolved. Taken together, these results suggest that the cross-sectional return structure is not stable. The characteristics that predict returns well in one decade may matter less in another, and treating predictor importance as fixed risks both overstating the durability of past findings and missing the rise of new drivers.

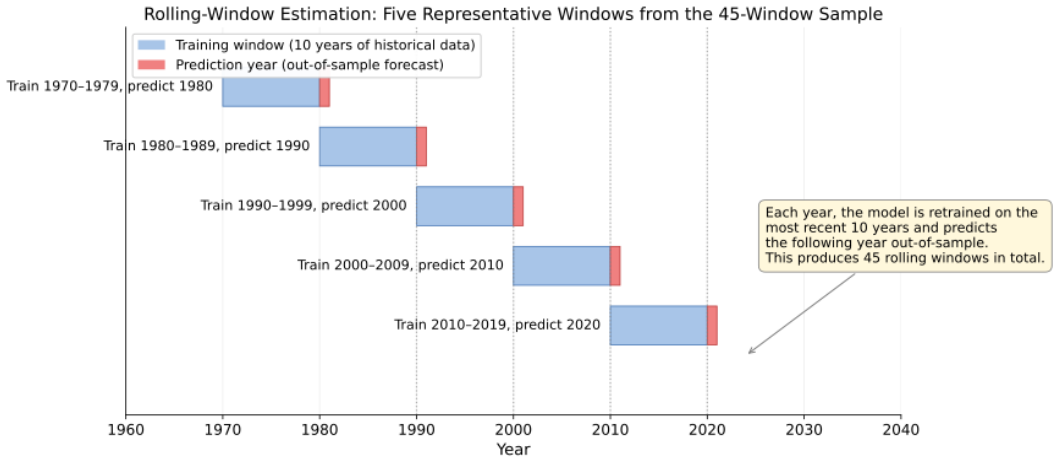
This thesis examines how the importance of firm-level characteristics as predictors of stock returns varies over time. The central research question is: *how do top firm-level characteristics identified by machine learning techniques as predictors of stock returns vary over time?* The question is fundamentally economic rather than methodological. The interest is not in benchmarking machine learning models against one another, but in using them as tools to track which firm characteristics are most informative about future returns in different historical periods, and whether the answer is stable across the past four decades.

Machine learning is well-suited to this task because it can simultaneously evaluate large numbers of predictors, capture nonlinear relationships, and apply disciplined out-of-sample validation. Gu et al. (2020) provide the influential reference point in this literature, demonstrating that machine learning methods generate substantial gains over linear factor models in predicting US stock returns when trained on a wide cross-section of firm characteristics. However, their analysis pools the entire 1957–2016 sample into a single expanding-window estimation, producing one set of importance rankings averaged across the full period. This thesis takes their framework as a starting point but reorients it toward time variation, replacing the expanding window with a fixed 10-year rolling window so that predictor importance can be tracked separately across decades.

The empirical analysis estimates three machine learning models, ElasticNet, Random Forest, and a feed-forward Neural Network, on a panel of US stocks covering 1970 to 2024, with the first ten years used to construct the initial training window and predictions generated for each year from 1980 to 2024. The estimation sample contains 1.28 million firm-month observations across 49 firm-level characteristics. For each model, the rolling 10-year training period generates a separate set of feature importance rankings for every prediction year, and hyperparameters are tuned independently per decade so that model complexity adapts to the period being estimated. Figure 1 illustrates this design, showing five representative windows from the full set of 45: each row corresponds to a single estimation window, with the training period in light blue and the prediction year in coral. The full set of 45 windows is presented in Section 4.3.

Figure 1

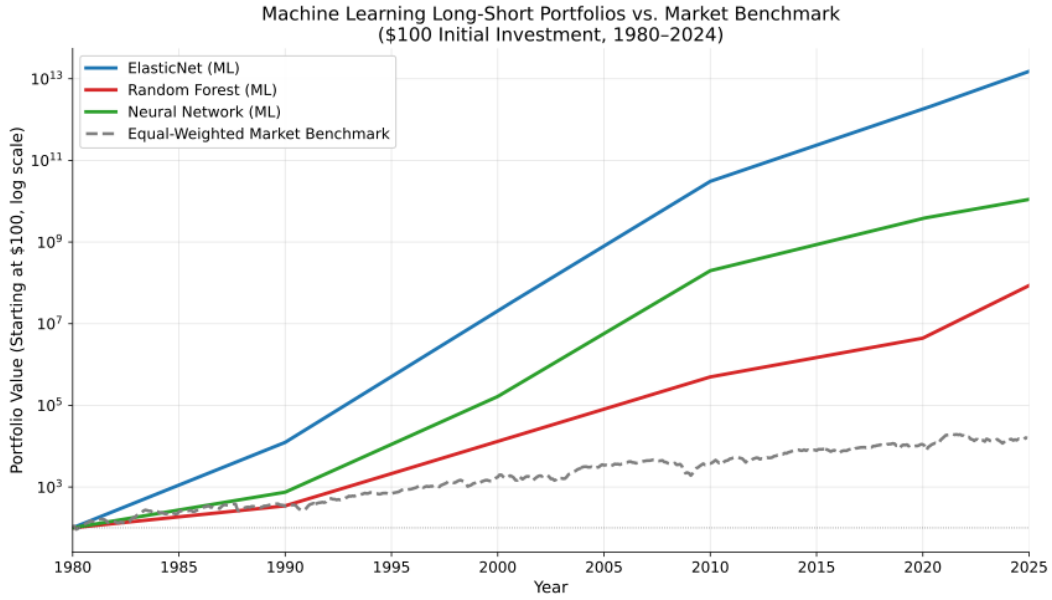
Rolling Window Illustration



Three main findings emerge. First, while size-related characteristics rank among the top predictors in every decade for all three models, the secondary predictors shift substantially over time. Momentum variables dominate the secondary tier of predictors in the 1980s and early 1990s but recede from the 2000s onward, consistent with the documented post-publication weakening of momentum-based anomalies. Liquidity measures gain importance in the 2000s, and accounting fundamentals, including the book-to-market ratio, return to prominence in the 2010s and 2020s. Second, an alpha-grid analysis tracking the order in which features enter the model under varying regularization confirms these shifts and reveals that the book-to-market ratio, after being outside the top ten earliest-entering features in the 1980s and 1990s, re-enters the top ten in the 2010s and 2020s. This nuances the popular "value investing is dead" narrative: the value premium has not disappeared but appears to have shifted in expression, consistent with recent literature. Third, long-short decile portfolios formed from the model predictions deliver economically meaningful Sharpe ratios across all three models and substantially outperform the equal-weighted market benchmark across the full sample period (Figure 2), though the magnitude and timing of returns differ across decades, reinforcing the conclusion that predictor importance is regime-dependent rather than stable.

Figure 2

Machine Learning vs. Market Benchmark Portfolio Value



This thesis contributes to the empirical asset pricing literature in three ways. It extends the Gu et al. (2020) framework by reorienting it toward time variation, replacing the expanding window with rolling estimation and introducing per-decade hyperparameter tuning. It documents systematic shifts in predictor importance across four decades, providing direct empirical evidence that the cross-sectional return structure evolves over time. And it complements the existing literature on individual anomaly decay, such as McLean and Pontiff (2016) and Hou et al. (2020), by analyzing decay and emergence patterns across a broad set of 49 characteristics simultaneously rather than one anomaly at a time.

The remainder of the thesis is organized as follows. Chapter 2 reviews the relevant literature on factor models, the factor zoo, machine learning in asset pricing, and time variation in return predictability. Chapter 3 describes the data and presents descriptive statistics. Chapter 4 details the empirical methodology, including the rolling window design, model specifications, hyperparameter tuning procedures, and feature importance metrics. Chapter 5 reviews the data quality and descriptive statistics. Chapter 6 presents the empirical results and analysis. Chapter 7 concludes.

2. Literature review

2.1. From Asset Pricing to Firm Characteristics

For decades, the value premium stood as one of the most robust findings in empirical asset pricing: stocks with high book-to-market equity consistently outperformed their low book-to-market counterparts (Fama & French, 1992, 1993). Yet over the past decade this premium has largely disappeared. Fama and French (2021) document that value premiums were substantially lower in the second half of their 1963–2019 sample, and Gonçalves and Leonard (2023) suggest part of this decline may be attributable to book equity becoming a weaker proxy for a firm's fundamental equity value, eroding the informational content of the traditional measure. This pattern raises a question that sits at the heart of this thesis: if one of the most established predictors of stock returns can lose its predictive power over time, which characteristics actually remain informative, and for how long?

To answer that question, it helps to understand where firm characteristics come from as predictors in the first place. Asset pricing theory establishes that expected returns should compensate investors for bearing risk that cannot be diversified away (Markowitz, 1952; Lintner, 1965; Sharpe, 1964). In the Capital Asset Pricing Model (CAPM), a stock's expected return is determined entirely by its sensitivity to the market portfolio (Lintner, 1965; Sharpe, 1964). However, later empirical evidence showed that market beta alone cannot fully account for cross-sectional variation in average stock returns (Black et al., 1972; Fama & French, 1992; Roll, 1977). This led to a shift toward multi-factor models in which observable firm-level characteristics serve as proxies for exposure to multiple priced risks. Size and book-to-market equity emerged as the canonical examples (Fama & French, 1992, 1993), followed by momentum (Jegadeesh & Titman, 1993; Carhart, 1997), and eventually a much broader set of characteristics such as profitability, investment intensity, and liquidity (Fama & French, 2015; Gu et al., 2020).

The size effect offers a useful point of contrast to the declining value premium. While the value premium has weakened substantially (Fama & French, 2021; Gonçalves & Leonard, 2023), size-related characteristics appear to be more persistent (Asness et al., 2018). Firm size captures relatively stable cross-sectional differences in liquidity, information environment, and trading frictions (Amihud, 2002; Daniel & Titman, 1997; Asness et al., 2018). This contrast between value and size implies that the predictive relevance of firm characteristics may differ substantially across time, making it unlikely that all characteristics remain equally informative over long sample periods. This observation directly motivates the research design of this thesis, which examines which firm characteristics emerge as important predictors across successive decades, rather than assuming a fixed set of dominant signals.

2.2. The Factor Zoo and the Selection Problem

The shift toward a characteristics-based perspective brought a practical problem: the number of proposed return predictors has grown enormously, to the point where Cochrane (2011) famously described the resulting collection as a "factor zoo." This expansion creates a high-dimensional selection problem that is directly relevant to this thesis. When hundreds of characteristics are tested across different sample periods, portfolio constructions, and control settings, the probability of false discoveries rises mechanically with the number of signals considered, making it increasingly difficult to distinguish genuine predictors from chance findings (Harvey et al., 2016; White, 2000; Sak et al., 2024). In the case of cross-sectional return predictors Harvey et al. (2016) formalize factor discovery as a multiple testing problem and argue that the standard significance cutoffs in asset pricing tests are too lax given the large number of factors and the inevitable data mining. Therefore, a credible selection of predictors should rely on stricter inference standards and robustness tests that specifically target false discoveries.

The replication literature makes clear that this concern is not merely theoretical. A study by McLean and Pontiff (2016) indicates that the long-short returns of many reported characteristics decline out-of-sample and decline further after publication, reflecting a combination of statistical overestimation at discovery and subsequent learning or arbitrage by investors. Hou et al. (2020) complement this evidence, they find that when using stricter portfolio construction standards, roughly 65% of the 452 anomalies in their sample fail to meet even the conventional significance level, and imposing a multiple-testing correction raises the failure rate to 82%. Overall, these findings suggest that a significant portion of the factor zoo consists of predictors that are either statistically fragile or economically arbitraged away shortly after discovery. For this thesis, the replication evidence has a direct implication: predictor importance should not be evaluated based on full-sample in-sample statistics alone, but rather on robustness and out-of-sample performance.

2.3. Why Time Variation Matters

The replication evidence in the previous section demonstrates that many predictors weaken out of sample, but it does not fully explain why. Understanding the sources of time variation in characteristic-based return premia is important for this thesis, as it influences what type of patterns to expect in the empirical results and how to interpret them.

One explanation comes from the economics of arbitrage. In frictionless markets, mispricings would be quickly corrected by rational investors. In practice, arbitrage requires capital and exposes investors to the risk that prices move further against them before correcting, which may force arbitrageurs to liquidate positions prematurely (Shleifer & Vishny, 1997). In addition to this view, De Long et al. (1990) demonstrate that noise trader risk can discourage rational traders from betting aggressively against mispricing, allowing price deviations from fundamentals to persist. Guidolin and Timmermann (2008) provide direct empirical evidence of this and document strong regime dependence in the size and value premia. Together, these mechanisms suggest that the predictive importance of characteristics is likely state-dependent rather than constant over the full sample. As a result, full-sample averages can be misleading because they may smooth out periods in which a characteristic is highly informative relative to other periods in which it is weak or

absent. Therefore, this thesis tracks predictor importance across rolling decades rather than averaging over the full sample.

A second and more subtle source of instability is measurement. Many firm characteristics are constructed from accounting variables whose definitions, recognition rules, and reporting environments change over time. Consequently, the same signal may not be perfectly comparable across decades. Major changes in accounting standards can mechanically shift common ratios: for example, new rules for the treatment of lease agreements have led to the recognition of large liabilities on the balance sheet, which had a significant impact on ratios based on leverage and profitability (Dyer et al., 2025). This matters for interpreting the results of this thesis: when the importance of an accounting-based characteristic shifts across decades, the shift may reflect not only a changing economic relationship but also a changing definition of the underlying variable. This interpretation is consistent with Linnainmaa and Roberts (2018), who show that most accounting-based anomalies weaken out of sample and are sensitive to sample choice, suggesting that temporal instability is central to understanding the cross-section of stock returns.

2.4. High-Dimensional Methods

The factor zoo problem and the time variation documented in the previous sections together define the empirical challenge this thesis faces: how to reliably identify which firm characteristics contribute most to return prediction in a high-dimensional setting where many signals are correlated, and their relative importance shifts over time. Three broad approaches have been developed in the literature to address this challenge, each with different strengths and limitations that are directly relevant to the design choices made.

The first approach uses shrinkage or compression to summarize the pricing information contained in many correlated predictors into a lower-dimensional representation. Kozak et al. (2020) show that a stochastic discount factor constructed from a small number of principal components performs well out of sample, and that shrinking the contribution of low-variance components to zero improves robustness by limiting overfitting. This approach is economically plausible because it treats pricing information as dispersed across many related signals rather than concentrated in a few isolated predictors. The main drawback for this thesis is interpretability: once the pricing object is constructed as a combination of many signals, it becomes difficult to attribute expected-return variation to specific firm characteristics, which is precisely what this thesis aims to do.

The second approach uses selection-based methods to assess whether a proposed factor provides incremental explanatory power after accounting for a large set of competing factors. Feng et al. (2020) show that simply applying a model-selection procedure such as LASSO and then testing the significance of a new factor can lead to unreliable inference in a high-dimensional setting. To obtain selection-robust inference, they propose a double-selection LASSO procedure designed to make inference on factor importance more robust to model-selection mistakes (Belloni et al., 2014; Chernozhukov et al., 2015). This approach is useful for evaluating whether individual characteristics hold up against a large benchmark set of competing factors, but its primary goal is inference on factor significance rather than tracking how predictive importance evolves over time.

The third approach, and the one most directly relevant to this thesis, uses machine learning methods capable of handling a large number of predictors simultaneously, capturing nonlinear relationships and interactions, and applying disciplined out-of-sample validation (Gu et al., 2020; Varian, 2014). Gu et al. (2020) provide the benchmark study in this literature, showing that neural networks and tree-based models generate substantial out-of-sample predictive gains over linear models, with much of the advantage coming from capturing nonlinear predictor interactions. Crucially, their framework treats empirical asset pricing as a prediction problem: models are evaluated on out-of-sample forecast performance and the economic value of the resulting portfolio strategies rather than on the interpretability of a small set of coefficients. This is the definition of importance adopted in this thesis.

Because many high-performing machine learning models function as black boxes, researchers often use post-hoc interpretation or feature-importance tools to assess which characteristics drive model forecasts (Guidotti et al., 2018). In tree-based models, permutation-based variable importance measures how predictive accuracy deteriorates when a feature is randomly permuted (Breiman, 2001). More recently, SHAP (SHapley Additive exPlanations) has offered a unified game-theoretic method that assigns an importance value to each feature for a given prediction, offering a transparent way to assess which characteristics drive machine learning forecasts (Lundberg & Lee, 2017).

The trade-off across these three approaches is clear. Shrinkage methods produce robust pricing objects but come at the expense of characteristic-level interpretability (Kozak et al., 2020). Selection-based methods such as Feng et al. (2020), enable rigorous inference on individual factors but they are not designed to track time variation. Machine learning methods offer the greatest predictive flexibility and can be combined with transparent importance metrics, but require disciplined out-of-sample evaluation to prevent overfitting. This thesis adopts the machine learning approach of Gu et al. (2020) as its primary framework, extends it toward time variation through rolling window estimation, and uses transparent importance metrics to track which characteristics matter most across decades.

2.5. Synthesizing the Evidence: What the Core Papers Imply for the Research Question

The previous sections establish three related aspects. First, firm characteristics differ in how persistently they predict returns: the contrast between the declining value premium and the more stable size effect indicates that not all predictors remain equally informative over extended sample periods. Second, many reported predictors are fragile, as demonstrated by the factor zoo and replication evidence, and their significance should be assessed out of sample rather than using full-sample statistics. Third, time variation in return premia arises from real economic mechanisms, including limits to arbitrage and changing accounting definitions, implying that the cross-sectional return structure is regime-dependent rather than stable.

These points indicate a gap in the existing literature. Gu et al. (2020) demonstrate that machine learning methods can extract substantial out-of-sample predictive power from a large set of firm characteristics, and their feature importance rankings offer a useful starting point. However, their analysis pools the full sample into a single expanding estimation

window, resulting in a single set of characteristic rankings averaged over the entire period. Consequently, their framework is unable to determine whether the characteristics that are most important in one decade are also the most important in another. Sak et al. (2024) take a step toward addressing this by showing how characteristic importance in machine learning models varies over an 18-year out-of-sample period. However, that evidence covers a relatively short period and does not systematically compare how importance rankings evolve across consecutive decades spanning the entire period after 1980. No study has examined how the ranking of top firm-level characteristics evolves across rolling windows spanning multiple decades within a consistent empirical framework. This thesis addresses that gap directly.

Therefore, the contribution of this thesis is not to claim for the first time that characteristic importance varies over time, but to examine that variation more systematically and over a longer horizon than existing work. Following Gu et al. (2020), a characteristic is defined as important if it improves out-of-sample return prediction, and importance is tracked separately across rolling 10-year windows as opposed to being averaged over the full sample. Throughout this thesis, two interpretive cautions from the literature are kept in mind. Kozak et al. (2020) warn that a characteristic may rank highly in one period because it proxies for broader information shared with correlated predictors rather than because it uniquely captures a distinct effect, so feature rankings should not be interpreted too mechanically. Freyberger et al. (2020) further caution that high predictive relevance does not automatically imply high incremental explanatory content.

Based on the evidence reviewed in sections 2.2 and 2.3, a plausible range of expectations for the empirical results can be stated. Price- and trading-based signals such as momentum and liquidity measures may show greater persistence in importance rankings because they are less affected by changes in accounting standards discussed in section 2.3 and because they capture market microstructure effects that are likely to remain relevant across regimes. Accounting-based characteristics, by contrast, may show stronger time-dependent patterns both because their measurement changes over time and because they are more vulnerable to post-publication arbitrage and sample sensitivity. However, because machine learning rankings can also shift when correlated predictors substitute for one another across subperiods, these expectations should be treated as hypotheses to be tested rather than inevitable conclusions.

3. Data

3.1. Data Sources

This research combines accounting information from Compustat with stock market data from CRSP (Center for Research in Security Prices), both accessed via the Wharton Research Data Services (WRDS) platform. These databases are widely regarded as the standard sources for empirical research in asset pricing and corporate finance.

Monthly stock returns and price information are obtained from the CRSP Monthly Stock File. Daily stock data from the CRSP Daily Stock File are used to construct realized volatility, maximum returns, and other intra-month measures that cannot be computed from monthly data alone. Annual accounting data are extracted from the Compustat North America Fundamentals Annual (FUNDA) dataset. The sample period spans 1970–2024 for CRSP data and 1969–2024 for Compustat data, providing over five decades of data to analyze time-varying predictive relationships. Compustat begins one year earlier to ensure accounting data is available with the required six-month lag when matched to 1970 stock returns.

All data processing, cleaning, and variable construction are conducted in Python 3.11.9, using the pandas and NumPy libraries for data manipulation and numerical operations, and PyArrow for efficient storage in parquet format.

For both CRSP and Compustat datasets, variable names were standardized to ensure consistent referencing throughout the analysis. In the case of Compustat, all column names were converted to lowercase. CRSP variables were used as exported from WRDS, following their standard uppercase naming convention. A complete overview of all variable definitions, construction formulas, and data source mnemonics is provided in Appendix A.

3.2. CRSP Data

3.2.1. Universe and Filters

To ensure a clean and investable sample, only common shares (CRSP share codes 10 and 11) listed on NYSE, AMEX, or NASDAQ (exchange codes 1, 2, and 3) are retained. Negative CRSP prices, which reflect bid-ask conventions, are converted to absolute values. Market equity is computed as the product of absolute price and shares outstanding. Following Gu et al. (2020), two additional screens are applied prior to model estimation: stocks below the 20th percentile of NYSE market capitalization each month are excluded, and stocks with a price below $1 are removed. These filters eliminate micro-cap stocks whose returns are disproportionately influenced by illiquidity and measurement error.

3.2.2. Return Construction

Monthly returns are adjusted to incorporate delisting returns following Shumway (1997), ensuring that returns reflect the full economic performance of firms that exit the sample and thereby avoiding survivorship bias. Missing delisting returns for performance-related delistings (delisting codes 500–584) are set to −30%. The full construction details are provided in Appendix A.

3.2.3. Predictor Variables

Firm size is defined as the logarithm of lagged market equity to avoid look-ahead bias. Lagged market equity in levels is retained as a separate predictor, as Gu et al. (2020) find that both the logarithmic and level specifications contain distinct predictive information. Short-term momentum is the lagged one-month return, identified as the single most important predictor in neural network models by Gu et al. (2020). Medium-term momentum is the cumulative return from month t−12 to t−2, excluding the most recent month to avoid short-term reversal effects. The change in momentum captures shifts in momentum trends as the difference between 6-month and 12-month momentum, where both measures are computed over lagged windows to avoid short-term reversal contamination.

Three liquidity measures are constructed: share turnover, dollar volume, and bid-ask spread. Dividend yield is approximated as the difference between total return and ex-dividend return, with missing values set to zero. Industry momentum is the equal-weighted average return across all firms within the same four-digit SIC industry code, lagged one month to avoid look-ahead bias.

Daily CRSP data are used to construct realized volatility, defined as the annualized standard deviation of daily returns within each month, requiring a minimum of 15 trading days. Maximum and minimum daily returns capture tail risk and lottery-like characteristics. Skewness and kurtosis of daily returns capture return asymmetry and tail thickness. For the liquidity measures, dollar volume, share turnover, and bid-ask spread, daily observations are averaged within each month to provide more stable estimates than monthly snapshots, and these daily-derived measures replace their monthly counterparts in the final dataset. All daily-derived variables, specifically realized volatility, maximum and minimum daily returns, skewness, kurtosis, share turnover, dollar volume, and bid-ask spread, are lagged by one month to prevent look-ahead bias, as they are computed from within-month information. Industry momentum is separately lagged by one month as it uses the current month's cross-sectional returns to predict the following month. Complete variable definitions and formulas are provided in Appendix A.

3.3. Compustat Data

3.3.1. Filters and Date Handling

Annual accounting data are obtained from Compustat Fundamentals Annual and restricted to industrial format, standardized format, consolidated financial statements, and US firms reporting in USD. These restrictions ensure homogeneity in accounting definitions and eliminate potential distortions from foreign reporting standards.

To avoid look-ahead bias, a merge date is constructed as six months after the fiscal year-end date, following the Fama and French (1993) convention which assumes that annual accounting data are publicly available no earlier than six months after the fiscal year-end. This merge date is used when linking Compustat data to CRSP observations, with a maximum tolerance window of 18 months to avoid matching on stale accounting data. Compustat data are linked to CRSP observations using 8-digit CUSIP identifiers via a backward-looking asof merge, which associates each monthly observation with the most recently available accounting record within the 18-month tolerance window.

3.3.2. Book Equity

Book equity is constructed following the standard Fama and French (1993) definition as stockholders' equity plus deferred taxes and investment tax credits minus preferred stock. The preferred stock measure follows a hierarchy of redemption value, liquidating value, and carrying value, set to zero if all are missing. If stockholders' equity is missing, common equity is used as a fallback. The full construction formula is provided in Appendix A.

3.3.3. Accounting Predictors

In total, 35 firm-level predictors are constructed from Compustat, capturing multiple economic dimensions: profitability, growth, investment, leverage, liquidity, accruals, and efficiency. All ratios are scaled by total assets or sales where appropriate, and a safe division function prevents division-by-zero distortions throughout. Growth rates are computed using explicitly lagged firm-level values to ensure correct panel alignment.

Profitability measures include return on assets, return on equity, profit margin, gross profitability, operating profitability, EBIT-to-assets, EBITDA-to-assets, cash flow to assets, and asset turnover. Investment and growth measures include asset growth, sales growth, capital expenditure intensity, growth in net PP&E, intangible asset growth, goodwill growth, inventory growth, and employment growth. R&D, advertising, and SG&A intensity are included as proxies for intangible investment. Leverage measures include total leverage, debt-to-assets, debt-to-equity, and net debt-to-assets. Accounting liquidity measures include the current ratio, quick ratio, cash ratio, and working capital to assets. Accruals are constructed from balance sheet components following the standard approach in the literature. Year-over-year changes in ROA, profit margin, leverage, and asset turnover capture dynamic shifts in firm performance. Complete formulas for all variables are provided in Appendix A.

3.4. Final Dataset

The final dataset consists of 1,282,465 firm-month observations covering 12,624 unique firms over the period 1970–2024, after applying the NYSE size screen and price filter following Gu et al. (2020). The dataset contains 49 predictor variables spanning market-based signals, daily-derived risk measures, and accounting fundamentals. Book-to-market ratio is computed post-merge as it requires both accounting and market data. For R&D intensity, advertising intensity, intangible asset growth, and goodwill growth, missing values are set to zero as their absence genuinely indicates zero expenditure. For SG&A intensity, missing values are retained as they may reflect different reporting practices rather than zero spending. Cash flow to assets has approximately 47% missing values concentrated in the pre-1988 period, as operating cash flows only became mandatory under SFAS 95 in 1988. The sample selection process is summarized in Table 1.

Table 1

Sample Reconciliation

Panel A: CRSP Monthly Sample

Step	Observations
Raw CRSP monthly observations (1970–2024)	4,575,741
After share code filter (common shares only, codes 10 and 11)	3,329,839
After exchange filter (NYSE, AMEX, NASDAQ only)	3,269,536
After market equity filter (ME > 0)	3,230,858
After dropping missing adjusted returns	3,230,858
Final CRSP monthly sample	**3,230,858**

Panel B: CRSP Daily Sample

Step	Observations
Raw CRSP daily observations (1970–2024)	94,771,020
After share code and exchange filters	68,243,401
After minimum 15 trading days per month filter	3,219,146 monthly aggregates
Final CRSP daily sample (monthly aggregates)	**3,219,146**

Panel C: Compustat Annual Sample

Step	Observations
Raw Compustat observations after accounting filters	442,683
Unique firms	**35,666**

Panel D: Merged Sample and Final Universe

Step	Observations
After CRSP–Compustat merge (left join on CRSP)	3,230,858
After NYSE 20th percentile market cap screen	1,282,842
After price screen (stock price ≥ $1)	1,282,465
Final estimation sample	**1,282,465**

Note. Panel A shows the construction of the CRSP monthly sample. Panel B shows the CRSP daily sample, which is aggregated to monthly frequency and merged with Panel A using PERMNO and year-month identifiers, with daily-derived features replacing their monthly counterparts where applicable. Panel C shows the Compustat annual sample, which is linked to the CRSP observations in Panel D using 8-digit CUSIP identifiers via a backward-looking asof merge with a maximum tolerance of 18 months, following Fama and French (1993). The merge is a left join on the CRSP monthly sample, preserving all CRSP observations regardless of Compustat availability. The NYSE 20th percentile market cap screen and the $1 price screen applied in Panel D follow Gu et al. (2020). The final estimation sample covers 12,624 unique firms over the period 1980–2024, as the first 10 years (1970–1979) are used exclusively as the initial training window.

4. Methodology

4.1. Empirical Framework

This study builds on the empirical framework of Gu et al. (2020), with two design choices motivated by the central research question. First, models are estimated on a fixed 10-year rolling window rather than an expanding window, allowing predictor importance to be tracked across subperiods. Second, hyperparameters are tuned independently within the rolling window structure rather than fixed at the values reported by Gu et al. (2020), so that each model can adapt to the predictive structure of the specific period in which it is estimated. Together, these choices place time variation at the centre of the empirical design, directly addressing the research question of whether the drivers of return predictability are stable or regime-dependent.

The empirical setting consists of a large firm-month panel built from CRSP and Compustat data covering the period 1969–2024, with 49 predictor variables spanning market-based signals, daily-derived risk measures, and accounting fundamentals. A six-month accounting lag is applied throughout to prevent look-ahead bias, following Fama and French (1993). The primary evaluation criterion is out-of-sample (OOS) predictive performance, assessed through the OOS R^2 metric and the economic performance of long-short decile portfolios. This prediction-oriented definition of importance follows Gu et al. (2020), who argue that a characteristic is important if it improves OOS return prediction and generates economically meaningful gains when forecasts are translated into portfolio decisions.

4.2. Data Preprocessing

All preprocessing is performed strictly inside each rolling window using only the training data distribution, ensuring that no information from future periods contaminates the model inputs. Applying preprocessing to the full sample before splitting into windows would constitute a form of look-ahead bias known as data leakage, which can artificially inflate predictive performance.

Winsorization. Returns and all predictor variables are winsorized cross-sectionally at the 1st and 99th percentile each month. Winsorization bounds are computed from the monthly cross-sectional distributions within the training window, and the same bounds are applied to the prediction data. This approach follows Gu et al. (2020) and prevents extreme outliers from distorting model estimation while preserving the cross-sectional variation that drives return predictability.

Rank normalization. Following Gu et al. (2020), all predictor variables are cross-sectionally rank-normalized to the interval [−0.5, 0.5] each month according to:

$$\tilde{x}_{i,t} = \frac{rank(x_{i,t}) - 1}{max\left(rank(x_{i,t})\right) - 1} - 0.5 \qquad (1).$$

As shown in equation (1), each variable is ranked within its monthly cross-section and mapped to [−0.5, 0.5]. For the prediction data, values are mapped to the training distribution using the empirical quantile function, ensuring that the normalization is based solely on information available at the time of prediction. This transformation reduces the influence of outliers beyond winsorization and makes the predictor distributions comparable across variables and time periods.

Missing value treatment. After rank normalization, remaining missing values are set to zero. Since rank normalization maps the cross-sectional distribution to [−0.5, 0.5], a value of zero corresponds to the center of the training distribution and represents a neutral signal, neither high nor low relative to the cross-section. This is a more principled treatment than mean imputation, which would be subject to look-ahead bias if computed on the full sample.

Feature scaling. For the neural network model, a StandardScaler is additionally applied after rank normalization to ensure numerical stability during gradient-based optimization. The scaler is fitted on the training data only and applied to the prediction data.

4.3. Rolling Window Design

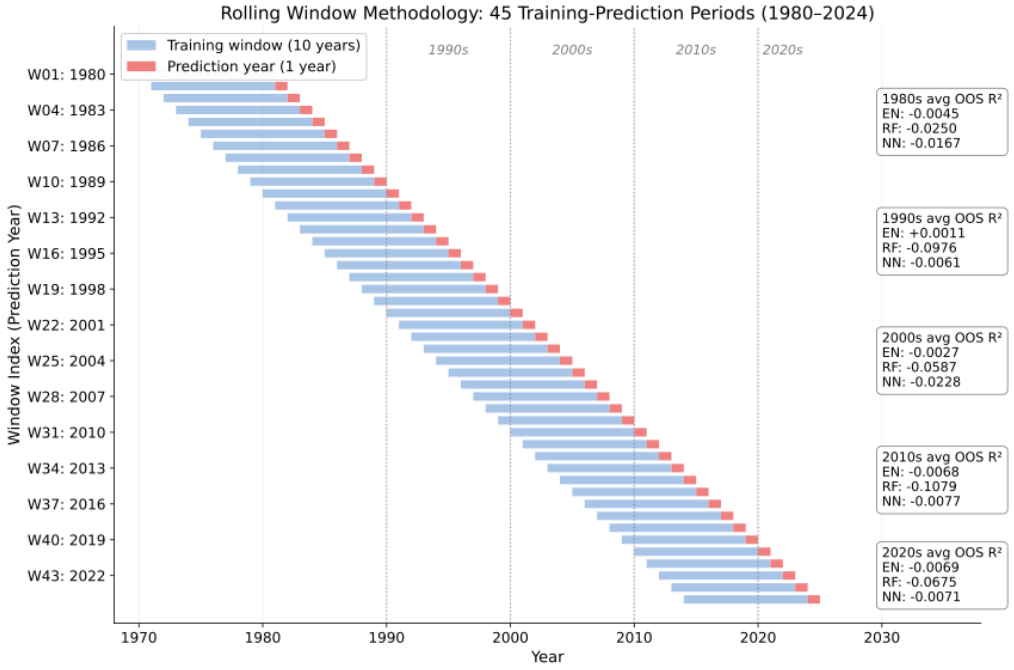
The rolling window design consists of a fixed 10-year training window followed by a 1-year prediction period, rolled forward annually. The first training window covers 1970–1979, with predictions generated for 1980. The window is then shifted forward by one year, training on 1971–1980 and predicting 1981, and so on. This produces 45 rolling windows in total, with prediction years spanning 1980–2024.

The choice of a rolling rather than expanding window follows directly from the research question. An expanding window, as used by Gu et al. (2020), continuously accumulates training data and is optimal when the relationship between predictors and returns is assumed to be stable over time. However, that assumption is precisely what this thesis tests. Pooling decades of data into a single training set forces the model to estimate one mapping from characteristics to returns and prevents direct comparison of predictor importance across subperiods. A rolling 10-year window instead restricts training to the most recent decade, allowing model parameters and the importance rankings they imply to evolve over time. This design follows Cooper et al. (2005) and Inoue et al. (2017), who demonstrate that rolling windows are appropriate when model parameters are expected to vary over time, which is the hypothesis motivating this study. Figure 3 visualizes the resulting design across all 45 windows.

Within each rolling window, all preprocessing steps described in Section 4.2. are applied independently. Models are retrained from scratch for each window, with hyperparameters selected through validation procedures specific to each model class, as detailed in Section 4.4. This ensures that all modeling decisions are made using only information available at the time of prediction, maintaining a fully out-of-sample evaluation design.

Figure 3

Rolling Window Methodology



Note. The figure shows all 45 rolling estimation windows used in this study. Light blue indicates the 10-year training period and coral indicates the 1-year prediction period for each window. Decade dividers are shown as vertical dotted lines, with the average out-of-sample R^2 for each model annotated on the right.

4.4. Machine Learning Models

Three machine learning models are estimated: ElasticNet regression, Random Forest, and a feed-forward Neural Network. This selection mirrors the model classes studied by Gu et al. (2020) and spans the interpretability-accuracy tradeoff discussed in the literature review. ElasticNet serves as the primary model due to its transparent coefficient structure, which directly enables the time-varying importance analysis central to this thesis. Random Forest and the Neural Network serve as comparison models that capture nonlinear relationships and interactions between predictors that ElasticNet cannot.

Hyperparameters are selected within the rolling window structure for all three models, rather than fixed at the values reported by Gu et al. (2020). The optimal degree of regularization or model complexity may differ across periods if the cross-sectional return structure itself evolves, and fixing hyperparameters globally risks suppressing precisely the time variation this study seeks to document. For the Random Forest and Neural Network, hyperparameters are tuned per decade rather than per window, a deliberate compromise between computational tractability and model adaptability that does not capture within-decade variation in optimal model complexity. ElasticNet, which is computationally cheaper, is retuned per window. The procedure differs across models and is described in detail below.

4.4.1. ElasticNet

ElasticNet regression combines L1 (LASSO) and L2 (Ridge) penalties in a single regularization framework. The objective function minimizes the mean squared prediction error subject to a penalty on the model coefficients:

$$\min_{\beta} \frac{1}{n} \sum_{i=1}^{n} (y_i - x_i'\beta)^2 + \alpha \left[\frac{1-\rho}{2} \|\beta\|_2^2 + \rho \|\beta\|_1 \right] \quad (2).$$

Equation (2) minimizes the mean squared prediction error subject to a combined penalty, where α controls the overall regularization strength and $\rho \in [0,1]$ controls the mix between Ridge ($\rho = 0$) and LASSO ($\rho = 1$). The L1 penalty induces sparsity by shrinking some coefficients exactly to zero, performing implicit variable selection. The L2 penalty handles correlated predictors more gracefully than pure LASSO by distributing coefficients across correlated features rather than arbitrarily selecting one. ElasticNet is therefore well-suited to a high-dimensional predictor set with correlated signals, as is common in empirical asset pricing.

Hyperparameters α and ρ are selected independently for each rolling window through 5-fold TimeSeriesSplit cross-validation. The grid considers values of α equal to 0.0001, 0.001, 0.01, 0.1, and 1.0, and values of ρ equal to 0.1, 0.5, 0.7, 0.9, and 1.0. TimeSeriesSplit is used rather than standard k-fold cross-validation to respect the temporal ordering of observations and prevent information leakage within the training window. The combination with the highest cross-validated R^2 is retained. Per-window tuning ensures that the regularization strength adapts to the predictive structure of each specific training period rather than being fixed at values calibrated on a different sample. Feature importance is quantified by the absolute magnitude of the estimated coefficients, which directly measures the contribution of each predictor to the linear prediction.

An alpha-grid analysis complements the coefficient-based importance metric. For each decade, a separate ElasticNet model is fit at each of 30 values of α spanning a logarithmic grid from 0.1 to 0.00001, with ρ fixed at 0.5. As α decreases, the regularization penalty weakens and additional predictors enter the model with non-zero coefficients. The order in which predictors enter measures relative importance: those that enter at higher α are selected first by the model and are therefore the most informative for that decade. For each combination of decade and α, the long-short portfolio Sharpe ratio is also computed, tracking performance as the model becomes progressively less sparse. Even when the top-ranked predictors are similar across decades, the order of entry can shift in ways the absolute coefficient ranking does not capture, and the relationship between performance and sparsity reveals whether predictability is concentrated in a few dominant features or distributed across many weaker signals.

4.4.2. Random Forest

Random Forest is an ensemble method that aggregates predictions from a large number of decorrelated regression trees. Each tree is trained on a bootstrap sample of the training data, and at each split only a random subset of $\sqrt{p}$ predictors is considered, where p is the total number of predictors. This double randomization, in observations and features,

reduces variance relative to a single tree while preserving the ability to capture nonlinear relationships and interactions between predictors.

The forest consists of 300 trees, a value fixed across all specifications since additional trees rarely improve predictive performance materially. The three remaining hyperparameters, namely the maximum tree depth, the minimum number of observations per leaf, and the number of predictors considered at each split, are tuned separately for each decade. Within the first rolling window of each decade, the ten-year training period is divided into the first eight years for training and the last two years for hyperparameter validation. The combination with the highest validation R^2 is retained and applied to all rolling windows within that decade. The grid considers maximum depths of 4, 6, 8, and 10, minimum leaf sizes of 500, 1,000, and 2,000, and either the square root of the number of predictors or one half of the predictors at each split, yielding 24 candidate combinations per decade.

A single train/validation split is preferred over k-fold cross-validation for the Random Forest and Neural Network because adjacent years within a 10-year training window often share market regimes, which causes k-fold validation to produce optimistic performance estimates and, for high-capacity nonlinear models, favor overly flexible specifications that fail to generalize to unseen years. The regularized linear structure of ElasticNet makes it substantially less vulnerable to this concern, which is why 5-fold TimeSeriesSplit is retained for that model. Tuning per decade rather than per window allows hyperparameters to adapt to broad shifts in the cross-sectional return structure while keeping the procedure computationally tractable.

Feature importance is quantified using permutation-based importance, which measures the decrease in OOS predictive accuracy when each predictor is randomly permuted in the prediction data. This approach, introduced by Breiman (2001), is preferred over the default impurity-based importance because it is evaluated out-of-sample and is therefore not subject to the in-sample bias that affects impurity measures for high-cardinality features.

4.4.3. Neural Network

The neural network follows the feed-forward architecture of Gu et al. (2020), consisting of three hidden layers with 32, 16, and 8 units respectively. Each hidden layer applies a linear transformation followed by batch normalization, a rectified linear unit (ReLU) activation function, and dropout regularization with a rate of 0.05. The output layer is a single linear neuron producing the return prediction. Batch normalization stabilizes training by normalizing layer inputs, while dropout prevents overfitting by randomly deactivating neurons during training.

The network is trained by minimizing the mean squared prediction error with L1 regularization on the network weights, following Gu et al. (2020). Optimization uses the Adam optimizer with a learning rate of 0.001, a ReduceLROnPlateau scheduler that reduces the learning rate by a factor of 0.1 when validation loss fails to improve for three consecutive epochs, and early stopping with a patience of 5 epochs that restores the weights from the epoch with the lowest validation loss. These optimization settings follow standard defaults rather than values tuned within this study; the tunable hyperparameters

are limited to the dropout rate and L1 regularization strength described below, in order to keep the computational burden of the rolling window estimation manageable.

The dropout rate and the L1 regularization strength are tuned separately for each decade using the same scheme as the Random Forest. Within the first rolling window of each decade, the last two years are held out for hyperparameter validation, the year preceding them is used for early stopping, and the remaining seven years are used for training. The grid considers dropout rates of 0.05 and 0.20 and L1 regularization strengths of 0.00001 and 0.0001, yielding four candidate combinations per decade. The combination with the highest validation R^2 is retained for all rolling windows in that decade. The architecture, batch size, and early stopping patience follow Gu et al. (2020), while the learning rate and ReduceLROnPlateau scheduler settings follow standard optimizer defaults. These are held fixed across all windows to limit the number of tunable parameters and reduce the computational burden.

Feature importance for the neural network is also quantified using permutation-based importance, consistent with the Random Forest approach, enabling direct comparison of feature rankings across models.

4.5. Portfolio Construction

Following Gu et al. (2020), portfolio performance is evaluated using equal-weighted decile portfolios sorted on predicted return each month. At the end of each month, all stocks in the investable universe are ranked by their predicted return for the following month. Stocks are assigned to one of ten deciles based on this ranking, with decile 1 (D1) containing the stocks with the lowest predicted returns and decile 10 (D10) containing those with the highest. The long-short portfolio goes long D10 and short D1, with monthly rebalancing.

The primary portfolio metric is the annualized Sharpe ratio, computed as:

$$SR = \frac{\sqrt{12} \cdot \bar{r}^e}{\sigma(r^e)} \tag{3}.$$

In equation (3), $\bar{r}^e$ is the mean monthly excess return and $\sigma(r^e)$ is its standard deviation, multiplied by $\sqrt{12}$ to annualize. Excess returns are computed relative to the risk-free rate obtained from the Ken French data library. Additional metrics reported include annualized return, annualized volatility, and maximum drawdown, defined as:

$$MDD = min_t \frac{C_t - max_{s \le t} C_s}{max_{s \le t} C_s} \tag{4}.$$

In equation (4), C_t is the cumulative portfolio return at time t, capturing the largest peak-to-trough decline over the sample period. Portfolio returns are computed using actual realized returns, not predicted returns, ensuring that portfolio performance reflects genuine economic gains rather than in-sample fit.

Actual returns used in portfolio construction are cross-sectionally winsorized at the 1st and 99th percentile each month to mitigate the influence of extreme individual stock returns on

equal-weighted portfolio averages, following standard practice in the empirical asset pricing literature.

4.6. Feature Importance Analysis

The primary contribution of this thesis is the analysis of how feature importance varies over time. For each rolling window, the importance of each predictor is quantified using the model-specific metric described in Section 4.4. Importance is then tracked across windows to examine whether the set of dominant predictors is stable or changes systematically across market regimes and decades.

For ElasticNet, the absolute coefficient magnitude serves as the importance measure. A non-zero coefficient indicates that the predictor is selected by the model in that window, while the magnitude reflects its contribution to the prediction. Tracking which predictors receive non-zero coefficients and their magnitudes across windows directly reveals how the linear return structure evolves over time.

For Random Forest and Neural Network, permutation-based importance scores are used. These scores are directly comparable across the two models since they are defined identically, as the increase in mean squared prediction error when a feature is randomly permuted in the out-of-sample data.

Decade-level summaries are constructed by averaging importance scores within each decade (1980s, 1990s, 2000s, 2010s, 2020s), providing an interpretable high-level view of how the most informative characteristics shift across distinct economic regimes. Cross-model comparison of importance rankings is used to identify predictors whose importance is robust across different modeling assumptions, a stronger form of evidence than importance established by a single model alone.

4.7. Evaluation Metrics

Out-of-sample R². The primary measure of predictive accuracy is the out-of-sample R², defined as:

$$R^2_{OOS} = 1 - \frac{\sum_{i,t}\left(r_{i,t} - \hat{r}_{i,t}\right)^2}{\sum_{i,t}\left(r_{i,t} - \bar{r}_t\right)^2} \quad (5).$$

In equation (5), $r_{i,t}$ is the realized return, $\hat{r}_{i,t}$ is the model's predicted return, and $\bar{r}_t$ is the cross-sectional mean return in month t. A positive OOS R² indicates that the model outperforms the historical mean return as a benchmark. This metric is preferred over in-sample R² because it directly measures generalization performance and is not inflated by overfitting. Gu et al. (2020) adopt the same metric as their primary evaluation criterion, facilitating direct comparison with their results.

Portfolio Sharpe ratio. While OOS R² measures statistical predictability, the Sharpe ratio of the long-short portfolio measures economic significance, whether the model's predictions

are useful enough to generate risk-adjusted profits after accounting for systematic risk. A model with modest OOS R^2 can still generate meaningful portfolio performance if its predictions are well-calibrated in the cross-section, as documented by Gu et al. (2020).

Decile return monotonicity. As an additional diagnostic, the mean monthly return of each decile portfolio (D1 through D10) is reported. A monotonically increasing pattern from D1 to D10 indicates that the model's predicted return ranking correctly orders stocks by their subsequent realized returns across the full cross-section, not just at the extremes. This provides stronger evidence of genuine predictive power than the long-short spread alone.

5. Data Quality and Descriptive Statistics

Prior to model estimation, the quality of the constructed dataset is validated to ensure that the return series and predictor variables are economically sensible. Table 2 reports descriptive statistics for the dependent variable and all 49 predictor variables, computed at the firm-month level across the full sample period.

The equal-weighted mean monthly return of 0.86% after winsorization is consistent with the long-run average documented in the empirical asset pricing literature. The return distribution exhibits positive skewness and excess kurtosis, reflecting the well-known fat tails of individual stock returns. Winsorization at the 1st and 99th percentile reduces the standard deviation from 18.96% to 15.79% and substantially reduces skewness from 9.02 to 1.16, confirming that extreme outliers are effectively controlled. The equal-weighted portfolio produces negative returns in all five major market downturns in the sample, October 1987 (−27.67%), August 1998 (−20.27%), September 2002 (−8.84%), October 2008 (−21.42%), and March 2020 (−23.00%), confirming that the return series correctly captures known market dislocations. The annualized Sharpe ratio of the equal-weighted universe is 0.30, which is economically plausible for a broad US equity portfolio over this period.

The stock universe follows the known CRSP composition pattern, growing from approximately 2,200 stocks in the early 1970s to a peak of around 7,500 in the late 1990s before declining to approximately 3,500–4,000 by 2024, as shown in Figure 4, consistent with the documented rise and subsequent consolidation of US public equity markets over this period.

Among the predictor variables, missing values are most prevalent for accounting-based measures, reflecting the coverage limitations of Compustat in the early sample period and the absence of certain line items for many firms. Notably, cash flow to assets exhibits approximately 47% missing values concentrated in the pre-1988 period due to the mandatory adoption of SFAS 95. Variables with structurally zero missing values, including R&D intensity, advertising intensity, intangible asset growth, and goodwill growth, reflect the zero-fill treatment applied where absence of reporting genuinely indicates zero expenditure.

Table 2

Descriptive Statistics

Panel A: Return Variable

Variable	Mean	Std	P1	P99	Skew	Kurt
RET_ADJ (raw)	0.0110	0.1896	-0.4000	0.5938	9.02	562.72
RET_ADJ (winsorized)	0.0086	0.1579	-0.3736	0.5245	1.16	7.51

Note. Panel A reports statistics for the dependent variable before and after cross-sectional winsorization at the 1st and 99th percentile. Missing values (%) are 0 for both rows.

Panel B: Predictor Variables

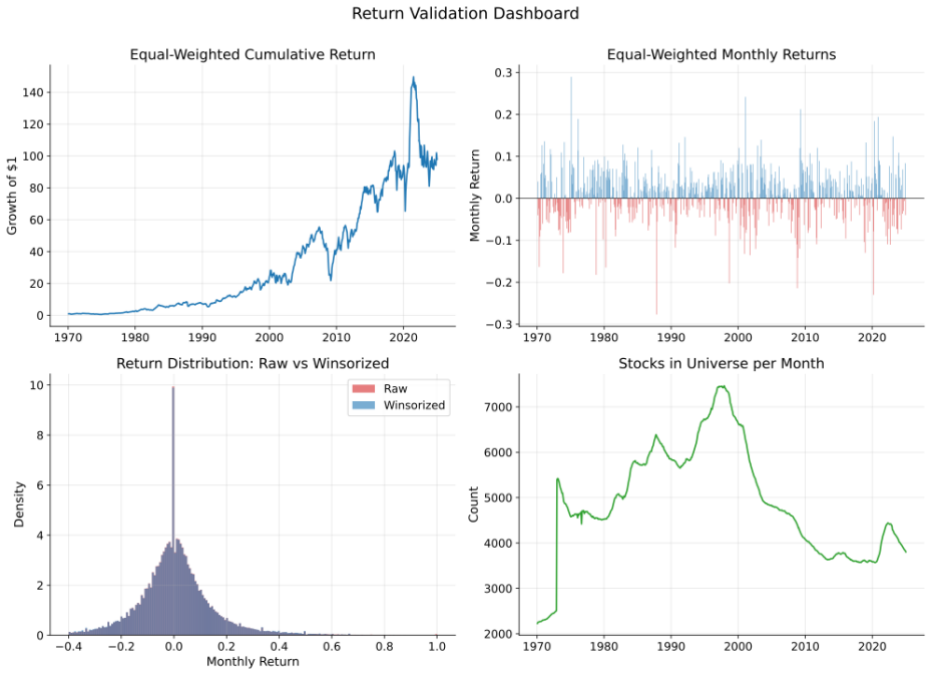
Variable	Mean	Std	P1	P99	Skew	Kurt
log_mvel1	18.5725	2.3041	14.0785	24.3386	0.34	-0.28
mvel1	1.83e9	9.14e9	1.30e6	3.72e10	11.71	175.56
mom1m	0.0088	0.1572	-0.3687	0.5217	1.16	7.45
MOM	0.1174	0.5997	-0.8216	2.2602	2.94	23.29
chmom	-0.0550	0.4202	-1.4839	0.8618	-2.33	24.26
turn	0.0057	0.0144	0.0001	0.0423	26.53	1270.64
dolvol	1.49e7	6.57e7	398.89	3.04e8	9.31	115.86
spread	0.0590	0.0724	0.0076	0.4000	4.52	28.13
dy	0.0011	0.0034	0.0000	0.0170	4.14	23.77
indmom	0.0105	0.0860	-0.2099	0.2585	0.55	4.90
retvol	0.5240	0.4126	0.0000	2.1447	2.32	8.18
maxret	0.0756	0.0739	0.0000	0.3813	2.95	13.44
minret	-0.0626	0.0521	-0.2653	0.0000	-2.15	6.42
skew_daily	0.2270	1.1179	-3.0837	3.3812	-0.13	3.17
kurt_daily	1.9366	3.5738	-1.1943	19.0000	2.75	9.42
bm	0.7952	0.9401	-0.4501	4.4351	4.66	57.10
roa	-0.0264	0.2383	-1.1554	0.2380	-4.25	25.74
roe	-0.0191	0.6992	-3.1775	1.6789	-2.77	34.32
profit_margin	-0.5377	5.2117	-13.2718	0.3726	-19.55	502.55
gross_profitability	0.3168	0.2974	-0.4526	1.2092	0.47	1.43
operating_profitability	0.0193	0.2258	-1.0191	0.3492	-3.39	17.01
ebit_to_assets	0.0187	0.2305	-1.0361	0.3512	-3.35	16.49
ebitda_to_assets	0.0578	0.2245	-0.9510	0.4003	-3.24	16.34
cashflow_to_assets	0.0127	0.2150	-0.9243	0.3387	-2.92	12.48
asset_turnover	1.0439	0.8782	0.0000	4.0893	1.42	3.20
asset_growth	0.2781	1.0974	-0.4463	4.3364	13.47	294.61
sales_growth	0.2355	0.7717	-0.6317	3.7096	7.37	77.18
capex_to_assets	0.0609	0.0716	0.0000	0.3571	2.59	9.02
ppent_growth	0.2831	0.9868	-0.5764	4.9581	7.08	71.58
intan_growth	0.1602	1.1618	-0.9203	5.4487	9.39	113.57
gdwl_growth	0.0628	0.5348	-0.6557	2.4889	9.51	114.47
invt_growth	0.2609	1.0331	-0.9928	5.0637	6.08	51.28
emp_growth	0.1057	0.3717	-0.5076	1.6800	4.24	29.93
rd_intensity	0.1821	1.8457	0.0000	3.7973	21.83	598.19
ad_intensity	0.0089	0.0266	0.0000	0.1396	6.58	72.33
sga_intensity	0.3929	0.9337	0.0295	3.9348	16.07	393.48
leverage	0.5516	0.2739	0.0554	1.3064	0.49	0.88
debt_to_assets	0.2309	0.2135	0.0000	0.9091	1.13	1.44
debt_to_equity	0.8690	2.0798	-3.9090	10.6893	2.63	24.70
net_debt_to_assets	0.0629	0.3468	-0.8784	0.8285	-0.49	0.44

current_ratio	2.9544	3.2326	0.3433	18.8832	4.16	24.01
quick_ratio	2.2927	3.1380	0.1849	18.0115	4.54	27.86
cash_ratio	1.3678	2.9921	0.0024	16.6528	4.77	29.37
working_capital_to_assets	0.2782	0.2555	-0.3380	0.8712	0.07	0.11
accruals	-0.0330	0.1013	-0.3722	0.2454	-0.62	4.40
delta_roa	0.0025	0.1848	-0.5969	0.7493	1.81	26.23
delta_margin	0.0507	2.2340	-2.9859	4.5802	6.61	415.54
delta_leverage	-0.0061	0.1569	-0.6314	0.4365	-2.52	24.13
delta_turnover	-0.0139	0.2756	-1.0456	0.7729	-1.16	7.66

Note. This table reports descriptive statistics for all 49 predictor variables, computed at the firm-month level over 1970–2024 on winsorized values (1st and 99th percentile cross-sectionally each month). intan_growth, gdwl_growth, rd_intensity, and ad_intensity have zero missing values, reflecting zero-fill treatment where absence of reporting indicates zero expenditure. cashflow_to_assets exhibits approximately 47% missing values concentrated in the pre-1988 period due to the mandatory adoption of SFAS 95. Most other accounting-based variables have 20–35% missing values reflecting limited Compustat coverage in the early sample period.

Figure 4

Return Validation Dashboard



Note. The four panels show diagnostic plots for the equal-weighted universe over the 1970–2024 sample period: cumulative return growth, monthly returns, the return distribution before and after winsorization at the 1st and 99th percentiles, and the number of stocks in the investable universe each month.

6. Results

6.1. Out-of-Sample Predictive Performance

Table 3 reports the out-of-sample R^2 for each model across all 45 rolling windows. The mean OOS R^2 is −0.004 for ElasticNet, −0.072 for Random Forest, and −0.013 for the Neural Network, with positive OOS R^2 in 53.3%, 24.4%, and 37.8% of windows respectively. While the negative mean values indicate that no model consistently outperforms the historical mean return as a cross-sectional benchmark, the results are nonetheless consistent with the broader empirical asset pricing literature. Gu et al. (2020) report OOS R^2 values of approximately 0.004 for linear models, and McLean and Pontiff (2016) document that return predictability weakens substantially after publication and over time, which is broadly consistent with the modest and time-varying predictability documented here.

Table 3

Out-of-Sample R^2 by Prediction Year

Year	ElasticNet	Random Forest	Neural Network
1980	-0.0291	-0.0150	-0.0276
1981	0.0119	-0.0471	-0.0282
1982	-0.0160	-0.0161	-0.0348
1983	-0.0133	-0.0094	-0.0074
1984	0.0088	-0.0484	0.0022
1985	-0.0287	-0.0244	-0.0127
1986	0.0155	-0.0865	0.0053
1987	0.0051	-0.0038	-0.0008
1988	0.0021	-0.0051	-0.0070
1989	-0.0012	0.0053	-0.0556
1990	-0.0066	-0.1626	-0.0528
1991	-0.0423	0.0599	-0.0298
1992	0.0127	-0.0130	0.0158
1993	0.0131	-0.2744	0.0098
1994	0.0143	-0.5919	0.0012
1995	-0.0230	-0.0031	-0.0149
1996	0.0075	-0.0006	-0.0105
1997	-0.0111	-0.0089	-0.0155
1998	0.0253	0.0035	0.0159
1999	0.0208	0.0151	0.0193
2000	0.0099	-0.0809	0.0234
2001	0.0324	0.0373	0.0398
2002	0.0021	-0.0114	0.0157
2003	-0.0282	-0.0204	-0.0077

2004	-0.0025	-0.0110	-0.0052
2005	0.0072	-0.0391	0.0018
2006	0.0049	-0.0145	-0.0058
2007	-0.0055	-0.0277	-0.0019
2008	-0.0330	-0.0504	-0.0369
2009	-0.0138	-0.3687	-0.2508
2010	-0.0165	0.0150	-0.0232
2011	0.0009	0.0049	0.0127
2012	0.0071	-0.0379	0.0078
2013	-0.0612	-0.0529	-0.0444
2014	0.0097	0.0088	0.0074
2015	0.0008	0.0116	-0.0103
2016	0.0013	-0.0134	0.0106
2017	-0.0015	-0.0769	-0.0010
2018	-0.0097	-0.3315	-0.0226
2019	0.0013	-0.6063	-0.0138
2020	0.0002	0.0210	0.0034
2021	-0.0041	-0.0221	-0.0052
2022	-0.0345	-0.1648	-0.0336
2023	-0.0001	0.0102	0.0028
2024	0.0041	-0.1820	-0.0028
Mean OOS R^2	**-0.0036**	**-0.0718**	**-0.0126**
% Positive Windows	**53.3%**	**24.4%**	**37.8%**

Note. OOS R^2 is computed relative to the cross-sectional mean return each month. Positive values indicate the model outperforms the historical mean benchmark. Rolling window trains on 10 years and predicts the following year. N = 45 windows (1980–2024).

Figure 5

Out-of-Sample R^2 by Prediction Year

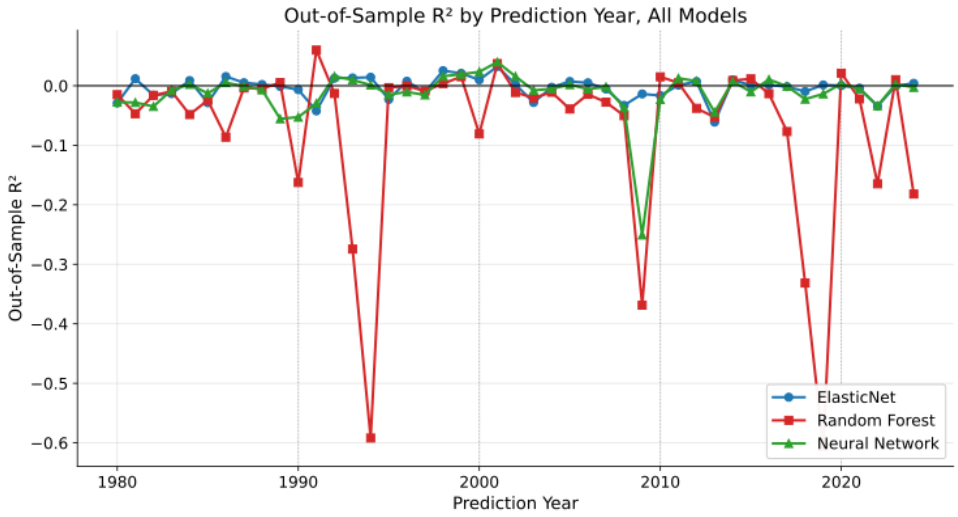


Figure 5 displays the OOS R^2 time series for all three models. ElasticNet is the most stable across decades, the Neural Network moderately variable, and the Random Forest substantially more volatile, with several windows producing values below −0.30. The greater instability of the Random Forest reflects its higher capacity to overfit when the cross-sectional return structure shifts between training and prediction periods.

Across all three models, predictability is highest during the late 1990s and early 2000s, with several windows producing OOS R^2 values above 0.02. Predictability deteriorates markedly during the 2008 financial crisis and again around 2018–2019, with multiple windows producing strongly negative OOS R^2 values for the nonlinear models. This pattern is consistent with the hypothesis that return predictability is regime-dependent and weakens during periods of heightened market uncertainty, when the cross-sectional return structure becomes more difficult to forecast. The partial recovery in predictability in the 2014–2024 period for ElasticNet and the Neural Network suggests that some degree of predictability persists in recent years, albeit at lower levels than in earlier decades.

ElasticNet outperforms both nonlinear models on OOS R^2 across all subperiods. This result is somewhat surprising given that Gu et al. (2020) find Neural Networks to be the best-performing model class. A likely explanation is the difference in window design: Gu et al. (2020) use an expanding window that continuously accumulates training data, whereas this study uses a fixed 10-year rolling window. Nonlinear models such as Random Forest and Neural Networks require larger training samples to reliably learn complex patterns, making them more susceptible to overfitting in shorter windows. ElasticNet, as a regularized linear model, is inherently more parsimonious and therefore better suited to the fixed rolling window design employed here.

6.2. Portfolio Performance

Table 4 reports portfolio performance metrics for the long-short decile strategy across all three models. ElasticNet generates the strongest risk-adjusted performance, with an annualized Sharpe ratio of 2.59 and a mean monthly long-short return of 4.7%. The Neural Network produces a Sharpe ratio of 2.12 and Random Forest 1.58, both economically meaningful but below ElasticNet. All three models generate positive long-short returns in every decade of the sample, confirming that the predictability captured is not concentrated in a single subperiod.

Table 4

Full Sample Portfolio Performance (1980-2024)

Panel A: ElasticNet

	Ann. Return	Ann. Vol.	Sharpe	Max DD
		ElasticNet		
Long (D10)	82.99%	29.72%	2.09	-27.91%
Short (D1)	6.02%	16.98%	0.20	-56.84%
Long-Short	74.10%	21.12%	2.59	-22.31%

Panel B: Random Forest

	Ann. Return	Ann. Vol.	Sharpe	Max DD
		Random Forest		
Long (D10)	54.65%	27.45%	1.60	-34.81%
Short (D1)	16.34%	19.20%	0.68	-42.79%
Long-Short	33.84%	16.88%	1.58	-21.56%

Panel C: Neural Network

	Ann. Return	Ann. Vol.	Sharpe	Max DD
		Neural Network		
Long (D10)	65.02%	28.25%	1.80	-27.25%
Short (D1)	11.05%	20.81%	0.42	-61.57%
Long-Short	49.14%	17.86%	2.12	-23.17%

Note. Long (D10) = top predicted-return decile; Short (D1) = bottom decile. Long-Short = zero-cost portfolio. Returns and volatility are annualized. Sharpe ratio is annualized (monthly Sharpe × √12). Max DD = maximum drawdown over full sample.

The decade-level performance is reported in Table 5 and visualized in Figure 6 below, which together reveal important time variation in portfolio returns. For ElasticNet, the long-short Sharpe ratio is highest in the 1980s (3.88) and 1990s (3.79), declines to 2.15 in the 2000s, partially recovers to 3.60 in the 2010s, and stands at 2.11 in the 2020s. The Random Forest and Neural Network show different decade patterns but similar overall magnitudes: the Neural Network peaks in the 2010s, while the Random Forest peaks in the 2020s. The high Sharpe ratios in the 1980s and 1990s are consistent with the documented proliferation of anomaly-based strategies in that period, before widespread awareness and arbitrage activity eroded returns. The recovery in the 2010s may reflect the emergence of new information environments, including high-frequency data and algorithmic trading, that created new cross-sectional dispersion in returns, partially offsetting the erosion of classical predictors.

Table 5

Decade-Level Long-Short Portfolio Performance

Panel A: ElasticNet

Decade	LS Monthly	LS Ann.	LS Sharpe	LS Max DD
		ElasticNet		
1980s	4.10%	61.14%	3.88	-2.13%
1990s	6.38%	106.85%	3.79	-4.36%
2000s	6.27%	98.57%	2.15	-22.31%
2010s	3.45%	49.27%	3.60	-5.96%
2020s	3.62%	50.58%	2.11	-12.28%

Panel B: Random Forest

Decade	LS Monthly	LS Ann.	LS Sharpe	LS Max DD
		Random Forest		
1980s	1.04%	12.80%	0.47	-11.61%
1990s	3.08%	42.10%	1.93	-11.39%
2000s	3.07%	39.76%	1.36	-21.56%
2010s	1.84%	23.92%	2.26	-8.55%
2020s	5.07%	78.81%	3.50	-5.27%

Panel C: Neural Network

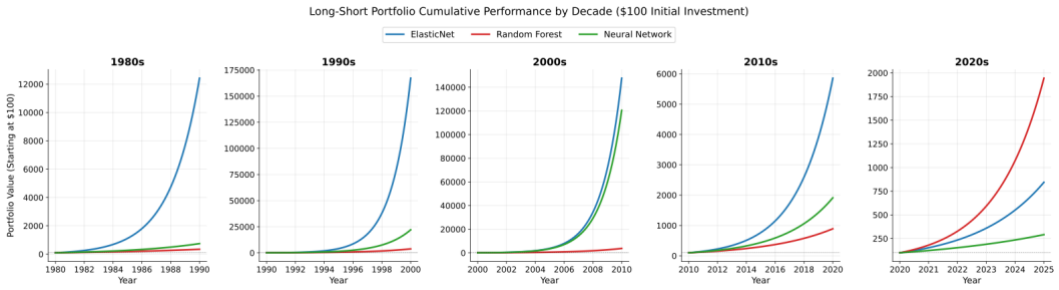
Decade	LS Monthly	LS Ann.	LS Sharpe	LS Max DD
		Neural Network		
1980s	1.69%	21.90%	1.37	-8.99%
1990s	4.60%	69.65%	3.08	-4.29%
2000s	6.09%	97.17%	2.57	-7.27%
2010s	2.49%	33.95%	3.92	-3.83%
2020s	1.80%	22.36%	1.20	-23.17%

Note. LS = Long-Short (D10 minus D1). LS Monthly = mean monthly long-short return. LS Ann. = annualized long-short return. LS Sharpe = annualized Sharpe ratio of long-short portfolio. LS Max DD = maximum drawdown within the decade.

The long leg (D10) consistently outperforms the short leg (D1) across all three models, suggesting the models are more effective at identifying outperformers than underperformers. This asymmetry is consistent with the literature on limits to arbitrage: short-selling constraints make it more difficult to profit from negative predictions, while long positions in high-predicted-return stocks can be implemented more freely. Long-short maximum drawdowns are −22.31% for ElasticNet, −21.56% for Random Forest, and −23.17% for the Neural Network, concentrated around the 2008 financial crisis and the COVID-19 period.

Figure 6

Long-Short Portfolio Cumulative Performance by Decade



6.3. Decile Return Monotonicity

Table 6 reports mean monthly returns for each predicted decile across all three models, with the corresponding patterns visualized in Figure 7. For ElasticNet, returns increase monotonically from 0.61% in D1 to 5.50% in D10, a pattern that strongly confirms the model's ability to rank stocks correctly across the full cross-section. The Neural Network also displays a monotonic pattern from 1.06% in D1 to 4.70% in D10. The Random Forest shows a different pattern: returns are roughly flat between D1 and D9 (1.34% to 1.82%) before jumping sharply to 4.00% in D10, suggesting that the model concentrates its predictive power in identifying extreme outperformers rather than ordering the full cross-section.

Table 6

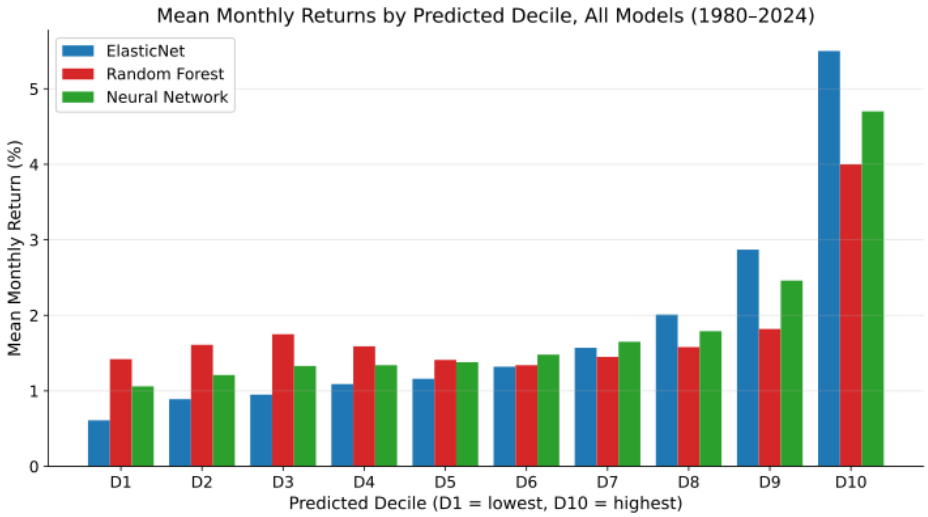
Mean Monthly Returns by Predicted Decile (1980–2024)

Decile	Mean Monthly	Ann. Return	Sharpe	Mean Monthly	Ann. Return	Sharpe	Mean Monthly	Ann. Return	Sharpe
	ElasticNet			*Random Forest*			*Neural Network*		
D1	0.61%	6.02%	0.20	1.42%	16.34%	0.68	1.06%	11.05%	0.42
D2	0.89%	9.63%	0.40	1.61%	18.68%	0.75	1.21%	13.67%	0.58
D3	0.95%	10.42%	0.44	1.75%	20.32%	0.78	1.33%	15.29%	0.67
D4	1.09%	12.21%	0.53	1.59%	18.65%	0.78	1.34%	15.39%	0.67
D5	1.16%	13.11%	0.57	1.41%	16.28%	0.70	1.38%	16.02%	0.70
D6	1.32%	14.98%	0.63	1.34%	15.28%	0.64	1.48%	17.36%	0.76
D7	1.57%	18.17%	0.75	1.45%	16.88%	0.73	1.65%	19.60%	0.85
D8	2.01%	24.24%	0.95	1.58%	18.40%	0.76	1.79%	21.20%	0.87
D9	2.87%	36.73%	1.28	1.82%	21.52%	0.85	2.46%	30.94%	1.18
D10	5.50%	82.99%	2.08	4.00%	54.65%	1.60	4.70%	65.02%	1.80

Note. D1 = lowest predicted-return decile; D10 = highest. Mean Monthly = average monthly portfolio return. Ann. Return = annualized return. Sharpe = annualized Sharpe ratio. N = 540 months for most deciles.

Figure 7

Mean Monthly Returns by Predicted Decile



The contrast in decile monotonicity across models is an important finding beyond what OOS R^2 captures. A model can have positive OOS R^2 while still failing to produce a well-ordered cross-section if its predictive power is concentrated in the tails. ElasticNet's clean monotonic pattern indicates that its linear predictions are well-calibrated across the full distribution of stocks, not only at the extremes. The Neural Network achieves a similar monotonicity, suggesting that the nonlinear flexibility of the network is captured in a way that still preserves cross-sectional ranking. The Random Forest, by contrast, behaves more like a binary classifier that separates a top decile from the rest, which is useful for long-only strategies but less informative for portfolio construction relying on intermediate deciles.

6.4. Time-Varying Feature Importance

Table 7 reports the top five predictors by decade for each model, representing the central contribution of this thesis. Across all three models and all decades, size-related variables, specifically log_mvel1 and mvel1, consistently rank among the most important predictors. This finding is consistent with the well-established size premium documented by Fama and French (1993) and suggests that firm size remains a robust and persistent predictor of cross-sectional returns throughout the full 1980–2024 sample period, even after applying the NYSE 20th percentile size screen that excludes the smallest stocks.

Table 7

Top 5 Predictors by Decade

Rank	ElasticNet	Random Forest	Neural Network
1980s			
1	log_mvel1	log_mvel1	profit_margin
2	dolvol	mvel1	roa
3	mom1m	rd_intensity	leverage
4	indmom	dy	debt_to_assets
5	minret	minret	asset_turnover
1990s			
1	log_mvel1	log_mvel1	mvel1
2	dolvol	mvel1	log_mvel1
3	turn	ad_intensity	leverage
4	retvol	spread	debt_to_equity
5	indmom	dy	ebit_to_assets
2000s			
1	log_mvel1	mvel1	log_mvel1
2	dolvol	log_mvel1	mvel1
3	turn	rd_intensity	debt_to_equity
4	maxret	leverage	profit_margin
5	spread	indmom	ad_intensity
2010s			
1	log_mvel1	mvel1	mvel1
2	dolvol	log_mvel1	dolvol
3	turn	leverage	log_mvel1
4	mvel1	gross_profitability	gross_profitability
5	rd_intensity	profit_margin	mom1m
2020s			
1	log_mvel1	mvel1	debt_to_assets
2	dolvol	rd_intensity	debt_to_equity
3	mvel1	spread	spread
4	turn	turn	roe
5	rd_intensity	retvol	leverage

Note. Feature importance for ElasticNet is based on mean absolute coefficient values averaged across all windows within the decade. For Random Forest and Neural Network, permutation importance is used. Rankings are computed separately for each decade.

Beyond size, the results reveal meaningful time variation in the importance of other predictor categories. For ElasticNet, the 1980s are characterized by momentum variables, particularly short-term momentum (mom1m) and industry momentum (indmom), consistent with the original momentum findings of Jegadeesh and Titman (1993). The 1990s shift toward liquidity measures, with dollar volume (dolvol) and share turnover (turn) gaining importance, coinciding with the rapid growth of institutional trading and the increasing role

of liquidity documented by Amihud (2002). From the 2000s onward, dollar volume and turnover dominate alongside size, while momentum-based predictors recede. This transition is consistent with McLean and Pontiff (2016), who document that momentum-based anomalies weaken after publication and after widespread adoption of momentum strategies by institutional investors. R&D intensity emerges as a top predictor in the 2010s and 2020s, suggesting that intangible investment has become an increasingly important driver of return variation in recent decades, consistent with Dyer et al. (2025).

The Random Forest and the Neural Network exhibit a different pattern. Both models also place size at or near the top of every decade, but the secondary predictors are distinct. For the Random Forest, R&D intensity is among the top predictors in three of the five decades, while for the Neural Network, profitability and leverage variables (profit_margin, roa, leverage, debt_to_assets) dominate the 1980s, with leverage and credit measures (debt_to_assets, debt_to_equity) becoming particularly prominent in the 2020s. The divergence between models suggests that the linear ElasticNet captures liquidity and momentum signals most effectively, while the nonlinear models capture interactions among accounting fundamentals that the linear specification cannot. The fact that all three models converge on size as a top predictor in every decade strengthens the case that size is a genuine and robust signal, while the divergence on secondary predictors reflects the differing inductive biases of linear versus nonlinear specifications.

The shift from momentum-dominated to liquidity-dominated importance rankings over time, particularly in the linear model, represents the central empirical finding of this thesis. It suggests that the cross-sectional return structure has fundamentally changed over the past four decades, with market microstructure and firm size becoming the primary drivers of predictability while traditional momentum effects have gradually eroded. This time-varying pattern is consistent with the theoretical predictions of Shleifer and Vishny (1997), who argue that the profitability of arbitrage strategies depends on market conditions and capital availability, implying that the dominant predictors should shift as market structure evolves.

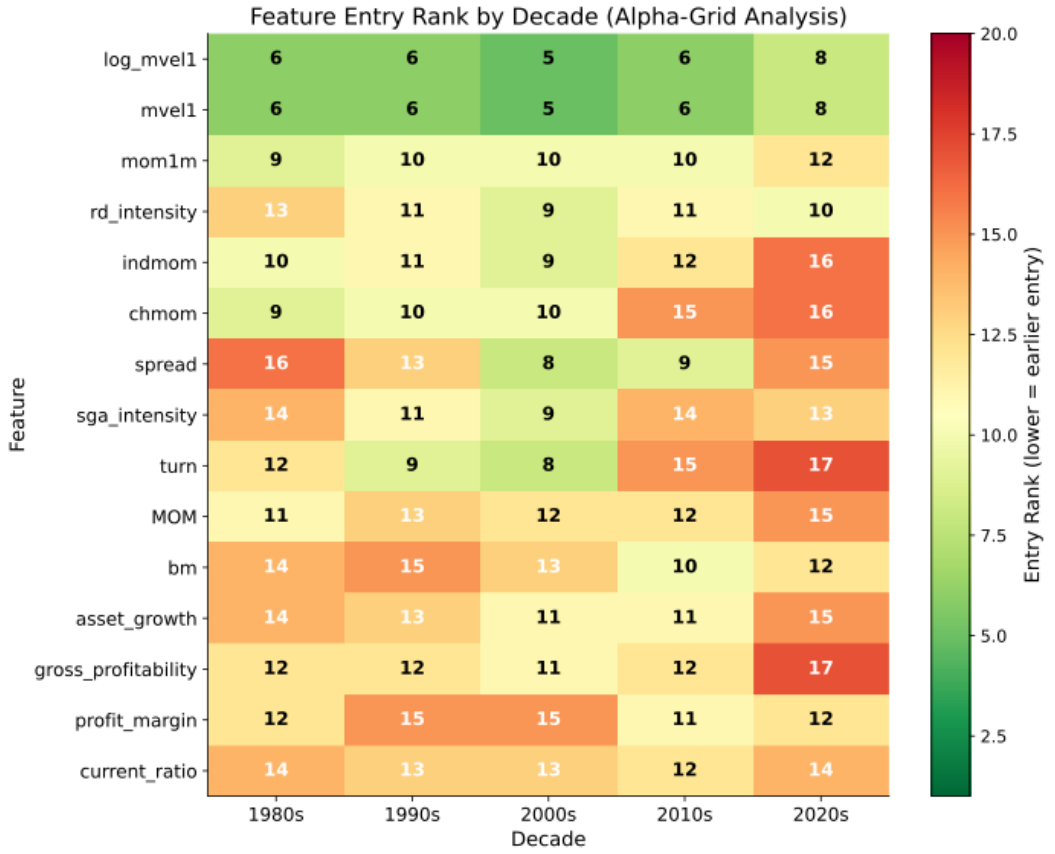
6.5. Alpha-Grid Analysis

The coefficient-based ranking in Table 7 identifies the predictors with the largest weight in each decade, but not the order in which features are selected as the regularization penalty is relaxed. Figure 8 reports the entry rank by decade for the fifteen most important features overall in the alpha-grid analysis, where lower entry ranks indicate earlier selection as α decreases. Log_mvel1 and mvel1 enter first in every decade, reinforcing the robustness of the size signal documented in Table 7. Beyond size, the entry order shifts meaningfully across decades. Momentum variables (chmom, indmom, MOM) rank in the top 10 throughout the 1980s and 1990s but drift to ranks 12 to 16 by the 2010s and 2020s, consistent with the documented weakening of momentum-based anomalies after their publication. Liquidity measures show distinct trajectories. Spread improves from rank 16 in the 1980s to ranks 8 and 9 in the 2000s and 2010s before receding to rank 15 in the 2020s. Turn rises from rank 12 in the 1980s to ranks 9 and 8 in the 1990s and 2000s, then recedes to rank 15 in the 2010s and rank 17 in the 2020s. The most striking pattern concerns the value indicator bm, which ranks 14 to 15 in the 1980s and 1990s, falls to rank 13 in the 2000s, and rises to rank 10 in the 2010s and rank 12 in the 2020s. This recovery is

consistent with recent literature suggesting that the value premium has not disappeared but has shifted in expression (Israel et al., 2021).

Figure 8

Feature Entry Rank by Decade

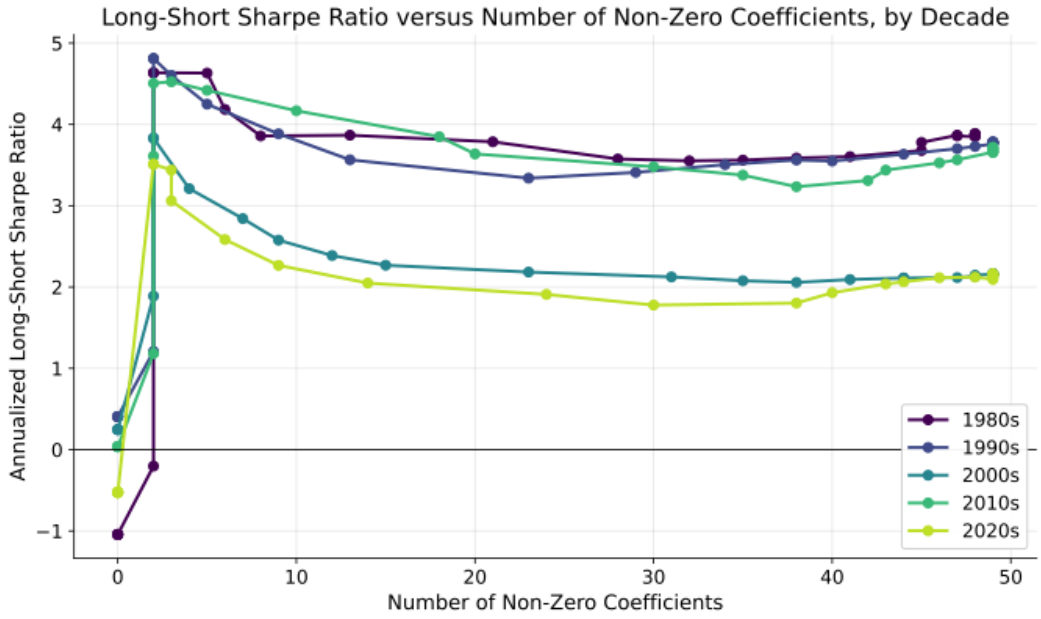


Note. The heatmap visualizes the entry rank of the 15 most important features across the five decades from the alpha-grid analysis. Lower ranks (greener cells) indicate that a feature is selected earlier as α decreases and is therefore more informative for that decade. Higher ranks (redder cells) indicate later entry. Numbers in each cell give the exact rank.

The relationship between portfolio Sharpe ratio and model sparsity, displayed in Figure 9, sheds further light on the structure of predictability. For each decade, the long-short Sharpe ratio rises sharply as the first few predictors enter the model and then stabilizes once roughly five to ten predictors are included. Beyond that point, adding additional predictors does not materially improve performance and in some decades slightly degrades it, consistent with the noise of less informative signals offsetting any marginal predictive content. This pattern indicates that predictability is concentrated in a small number of dominant features rather than distributed across many weaker signals. The 2020s display a flatter relationship between sparsity and Sharpe than earlier decades, consistent with weaker overall predictive signal in recent years.

Figure 9

Long-Short Sharpe Ratio versus Number of Non-Zero Coefficients



Note. Each line plots the annualized Sharpe ratio of the ElasticNet long-short decile portfolio as the number of non-zero coefficients varies, computed separately for each decade.

Taken together, these results refine the time-variation evidence in Table 7. Even when the top five predictors appear broadly stable across decades, the order in which predictors enter the model shifts substantially: momentum recedes from the 2000s onward, value re-emerges in the 2010s and 2020s, and accounting fundamentals gain prominence in the most recent decades. The marginal contribution of features therefore varies more than the absolute coefficient ranking alone would suggest, providing a more granular characterization of how the cross-sectional return structure has evolved.

6.6. Cross-Model Consistency

Reporting results across three model classes allows identification of predictors whose importance is robust to different modeling assumptions. Log_mvel1 and mvel1 appear in the top five predictors in every decade for all three models, providing strong cross-model evidence that firm size is a persistent and robust predictor of cross-sectional returns. Beyond size, the three models emphasize different secondary predictors: ElasticNet highlights liquidity (dolvol, turn) and momentum, the Random Forest highlights R&D intensity and liquidity, and the Neural Network highlights profitability and leverage. This divergence reflects the differing inductive biases of linear and nonlinear models: ElasticNet captures linear signals most effectively, while the Random Forest and Neural Network capture nonlinear interactions among accounting fundamentals. The agreement across models on size, alongside model-specific divergence on other predictors, strengthens rather than weakens the time-varying importance analysis: it indicates that the identified time variation reflects genuine shifts in the cross-sectional return structure rather than artifacts of a particular modeling assumption.

7. General Conclusion

This thesis examined how the importance of firm-level characteristics as predictors of cross-sectional stock returns has changed over the past four decades. Starting from the observation that classical predictors such as value and momentum have weakened substantially in recent years, while size and liquidity-based signals appear more persistent, the central question was whether the cross-sectional return structure is stable over time, or whether the characteristics that matter most for prediction shift across decades. Using a rolling 10-year estimation window applied to three machine learning models, ElasticNet, Random Forest, and a feed-forward Neural Network, on a panel of US stocks covering 1970 to 2024 with predictions generated for each year from 1980 to 2024, the analysis traced predictor importance separately for each prediction year and decade. The estimation sample contains 1.28 million firm-month observations across 49 firm-level characteristics. The results provide direct empirical evidence that predictor importance is regime-dependent, not stable, and that the cross-sectional return structure has evolved meaningfully across the sample period.

The central finding is that while size-related characteristics are persistently among the top predictors in every decade for all three models, the secondary predictors shift substantially over time. In the 1980s, momentum variables dominate the secondary tier, with short-term momentum, industry momentum, and 12-month momentum ranking among the most important features in both ElasticNet and the alpha-grid analysis. This is consistent with the original momentum findings of Jegadeesh and Titman (1993) and reflects an environment in which momentum-based strategies were not yet widely implemented by institutional investors. By the 1990s, the picture begins to change: turnover and dollar volume rise to the second tier of predictors, momentum loses ground, and intensity-based variables such as R&D and advertising intensity start appearing in the top rankings. This transition coincides with the rapid growth of institutional trading and the increasing role of liquidity, as documented by Amihud (2002).

The 2000s mark a more decisive shift. Liquidity measures such as bid-ask spread and turnover move firmly into the second tier of predictors across all three models, while pure momentum variables drift downward in importance. This pattern is consistent with McLean and Pontiff (2016), who document that momentum-based anomalies weaken systematically after publication and after widespread adoption by institutional investors. The decade also includes the 2008 financial crisis, during which predictability deteriorates sharply across all three models, suggesting that the cross-sectional return structure becomes harder to forecast during periods of heightened market uncertainty. The 2010s continue this evolution: liquidity remains important, R&D intensity rises further in the rankings, and accounting fundamentals such as profit margin, asset growth, and book-to-market begin re-entering the second tier of predictors. By the 2020s, the picture has changed substantially relative to the 1980s. Momentum variables, which dominated the secondary tier four decades earlier, no longer appear in the top ten earliest-entering features for ElasticNet. Instead, fundamentals such as book-to-market, profit margin, and EBITDA-to-assets enter notably earlier, and the alpha-grid analysis shows that the book-to-market ratio rises from rank 14–15 in the 1980s and 1990s to rank 10 in the 2010s and rank 12 in the 2020s. This recovery is consistent with recent literature suggesting that the value premium has not disappeared but has shifted in expression (Israel et al., 2021).

These per-decade patterns must be interpreted alongside an important caveat. The three models differ meaningfully in which secondary predictors they emphasize: ElasticNet highlights liquidity and momentum, the Random Forest emphasizes R&D intensity and leverage, and the Neural Network places weight on profitability and balance-sheet variables. This raises a legitimate question about whether the time variation documented in this study reflects genuine shifts in the underlying return structure, or merely artifacts of how different models capture and rank features. Three pieces of evidence support the former interpretation. First, all three models converge on size-related characteristics in every decade, providing strong cross-model evidence that size is a robust and persistent predictor. Second, the time variation in predictor importance is documented within each model in isolation, not just by comparing across models, which means that the same model trained on different decades produces materially different rankings. Third, the alpha-grid analysis, which uses a single fixed model class and varies only the regularization strength, also reveals systematic decade-level shifts in feature entry order, including the recovery of book-to-market and the decline of momentum. These findings indicate that while the choice of model influences which predictors rise to the top of any given ranking, the underlying time variation across decades is real and not driven purely by model choice or hyperparameter tuning.

The findings carry three implications for academic research. They support the view that anomaly-based asset pricing should be evaluated dynamically rather than over full samples, since static rankings can smooth over substantial decade-level variation in predictor importance. They reinforce the literature documenting post-publication anomaly decay and demonstrate that this pattern extends beyond individual factors to the broader structure of cross-sectional return predictability. And they highlight the value of reorienting the Gu et al. (2020) framework toward time variation by replacing the expanding window with rolling estimation, which provides a methodological template for future work on regime-dependent return predictability.

For practitioners, the implications are direct. A quantitative strategy calibrated on a long historical sample may underweight characteristics that have become more important recently and overweight characteristics whose informational content has decayed. The results suggest that practitioners should re-estimate factor importance periodically rather than treat predictor weights as time-invariant, and that the most informative characteristics for current return prediction may differ substantially from those identified by earlier academic literature.

The thesis has several limitations that open avenues for future research. The prediction sample begins in 1980, which excludes earlier decades when value and momentum effects may have been even stronger and prevents direct testing of the long-run evolution of these effects. The analysis is restricted to US equities, where the documented patterns may differ from those in international markets. Hyperparameters for the Random Forest and Neural Network are tuned per decade rather than per window, which is a deliberate compromise between computational tractability and model adaptability but means that the within-decade variation in optimal model complexity is not captured. The set of 49 predictors, while broad, is smaller than the several hundred characteristics studied by some recent papers in the factor zoo literature, and additional features could potentially reveal predictive structure not captured here. Finally, the analysis does not account for transaction costs in the long-short portfolios, which would compress the realized returns reported in the results section, particularly for the higher-turnover strategies.

These limitations suggest several directions for further research. Extending the framework to a longer historical sample using earlier CRSP data or to international markets would clarify whether the patterns documented here are specific to US equities or generalize globally. Tuning hyperparameters per window rather than per decade would allow the analysis to capture finer-grained variation in optimal model complexity. Expanding the predictor set to include text-based and alternative-data signals would extend the analysis to a richer information environment, particularly relevant for the most recent decade. And explicitly modeling regime transitions, rather than averaging within fixed decade boundaries, could provide a sharper characterization of when and how the cross-sectional return structure shifts.

The broader takeaway is that the question of which firm characteristics predict stock returns does not have a single time-invariant answer. The evidence presented here shows that the most informative characteristics shift across decades in patterns consistent with the evolution of market structure, the diffusion of academic research into investment practice, and the underlying changes in the cross-sectional return generating process. Treating predictor importance as fixed misses these shifts; treating it as dynamic, as this thesis does, reveals them.

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11. Appendix A

Variable Definitions and Construction

All variables are constructed from CRSP (monthly and daily stock files) and Compustat North America Fundamentals Annual, accessed via WRDS. The sample covers 1969–2024. Accounting data are matched to monthly stock returns using a six-month lag following Fama and French (1993). All daily-derived variables are lagged by one month to prevent look-ahead bias. A safe division function prevents division-by-zero distortions throughout.

Table 8

Predictor Variables

Variable	Description	Formula
log_mvel1	Log of lagged market equity	$ln(ME_{t-1})$
mvel1	Lagged market equity in levels	$ME_{t-1} = \|\|PRC\|\| \times SHROUT \times 1000$
mom1m	Short-term momentum (1-month lagged return)	RET_ADJ_{t-1}
MOM	Medium-term momentum (12-2-month cumulative return)	$\prod_{s=t-12}^{t-2}(1 + RET_ADJ_s) - 1$
chmom	Change in momentum (6m minus 12m momentum)	$mom6m_t - MOM_t$, where $\text{mom6m} = \prod_{s=t-6}^{t-2}(1 + RET_ADJ_s) - 1$
turn	Share turnover, daily average lagged 1 month	$VOL/(SHROUT \times 1000)$
dolvol	Dollar trading volume, daily average lagged 1 month	$\|\|PRC\|\| \times VOL$
spread	Bid-ask spread, daily average lagged 1 month	$(ASKHI - BIDLO)/[(ASKHI + BIDLO)/2]$
dy	Dividend yield	$RET - RETX$, missing set to 0
indmom	Industry momentum	Equal-weighted avg. return of same 4-digit SIC industry, lagged 1 month
retvol	Realized volatility, lagged 1 month	$\sigma^{daily} \times \sqrt{252}$
maxret	Maximum daily return, lagged 1 month	$max_{s \in t-1}(RET_s^{daily})$
minret	Minimum daily return, lagged 1 month	$min_{s \in t-1}(RET_s^{daily})$
skew_daily	Skewness of daily returns, lagged 1 month	$Skewness(RET_s^{daily}, s \in t-1)$
kurt_daily	Excess kurtosis of daily returns, lagged 1 month	$Kurtosis(RET_s^{daily}, s \in t-1)$
bm	Book-to-market ratio	$(BE \times 1{,}000{,}000)/ME_{t-1}$
roa	Return on assets	IB/AT
roe	Return on equity	IB/SEQ
profit_margin	Net profit margin	$NI/SALE$

gross_profitability	Gross profitability	$(REVT - COGS)/AT$
operating_profitability	Operating profitability	$OIADP/AT$
ebit_to_assets	EBIT scaled by total assets	$EBIT/AT$
ebitda_to_assets	EBITDA scaled by total assets	$EBITDA/AT$
cashflow_to_assets	Operating cash flow scaled by total assets	$OANCF/AT$
asset_turnover	Asset turnover ratio	$SALE/AT$
asset_growth	Annual asset growth rate	$(AT_t - AT_{t-1})/AT_{t-1}$
sales_growth	Annual sales growth rate	$(SALE_t - SALE_{t-1})/SALE_{t-1}$
capex_to_assets	Capital expenditure intensity	$CAPX/AT$
ppent_growth	Growth in net PP&E	$(PPENT_t - PPENT_{t-1})/PPENT_{t-1}$
intan_growth	Growth in intangible assets, missing set to 0	$(INTAN_t - INTAN_{t-1})/INTAN_{t-1}$
gdwl_growth	Growth in goodwill, missing set to 0	$(GDWL_t - GDWL_{t-1})/GDWL_{t-1}$
invt_growth	Growth in inventories	$(INVT_t - INVT_{t-1})/INVT_{t-1}$
emp_growth	Growth in number of employees	$(EMP_t - EMP_{t-1})/EMP_{t-1}$
rd_intensity	R&D intensity, missing set to 0	$XRD/SALE$
ad_intensity	Advertising intensity, missing set to 0	$XAD/SALE$
sga_intensity	SG&A intensity	$XSGA/SALE$
leverage	Total leverage ratio	LT/AT
debt_to_assets	Debt-to-assets ratio	$(DLC + DLTT)/AT$
debt_to_equity	Debt-to-equity ratio	$(DLC + DLTT)/SEQ$
net_debt_to_assets	Net debt-to-assets ratio	$(DLC + DLTT + CHE)/AT$
current_ratio	Current ratio	ACT/LCT
quick_ratio	Quick ratio	$(ACT - INVT)/LCT$
cash_ratio	Cash ratio	CHE/LCT
working_capital_to_assets	Working capital scaled by total assets	$(ACT - LCT)/AT$
accruals	Balance sheet accruals	$(\Delta WC_t - DP_t)/AT_t$ where $WC = (ACT - CHE) - (LCT - DLC)$
delta_roa	Year-over-year change in ROA	$roa_t - roa_{t-1}$
delta_margin	Year-over-year change in profit margin	$profit_margin_t - profit_margin_{t-1}$
delta_leverage	Year-over-year change in leverage	$leverage_t - leverage_{t-1}$
delta_turnover	Year-over-year change in asset turnover	$asset_turnover_t - asset_turnover_{t-1}$

Book Equity Construction

Book equity follows the standard Fama and French (1993) definition:

$$BE = SEQ + TXDITC - PS \quad (6).$$

where SEQ is stockholders' equity (with CEQ as fallback if missing), TXDITC is deferred taxes and investment tax credits, and preferred stock PS is defined hierarchically as PSTKRV >PSTKL >PSTK, set to zero if all are missing.

Return Construction

Monthly returns are adjusted for delisting following Shumway (1997):

$$RET_ADJ_{i,t} = \left(1 + RET_{i,t}\right)\left(1 + DLRET_ADJ_{i,t}\right) - 1 \quad (7).$$

where DLRET_ADJ equals the CRSP delisting return if available, -30% for performance-related delistings (codes 500–584) with missing delisting returns, and 0% otherwise.

Table 9

CRSP Variable Mnemonics

Mnemonic	Description
PERMNO	Permanent stock identifier — primary CRSP firm identifier
CUSIP	Committee on Uniform Security Identification Procedures identifier, used to match CRSP to Compustat
SICCD	Standard Industrial Classification Code
EXCHCD	Exchange code (1 = NYSE, 2 = AMEX, 3 = NASDAQ)
SHRCD	Share code (10, 11 = common shares retained in sample)
PRC	Monthly closing price (negative values indicate bid-ask average, converted to absolute value)
SHROUT	Shares outstanding in thousands
VOL	Monthly trading volume in number of shares
RET	Monthly holding period return
RETX	Monthly return excluding dividends
DLRET	Delisting return
DLSTCD	Delisting code (500–584 = performance-related delistings)
ASKHI	Highest ask price during the month
BIDLO	Lowest bid price during the month

Table 10

Compustat Variable Mnemonics

Mnemonic	Description
GVKEY	Global Company Key — primary Compustat firm identifier
CUSIP	Committee on Uniform Security Identification Procedures identifier, used to match Compustat to CRSP
DATADATE	Fiscal year-end date
FYEAR	Fiscal year
AT	Total assets
SALE	Net sales
REVT	Total revenues
COGS	Cost of goods sold
IB	Income before extraordinary items
NI	Net income
OIADP	Operating income after depreciation
EBIT	Earnings before interest and taxes
EBITDA	Earnings before interest, taxes, depreciation and amortization
OANCF	Operating activities — net cash flow
CAPX	Capital expenditures
XRD	Research and development expense
XAD	Advertising expense
XSGA	Selling, general and administrative expense
SEQ	Stockholders' equity
CEQ	Common/ordinary equity
TXDITC	Deferred taxes and investment tax credit
PSTKRV	Preferred stock redemption value
PSTKL	Preferred stock liquidating value
PSTK	Preferred stock carrying value
LT	Total liabilities
DLC	Debt in current liabilities
DLTT	Long-term debt
CHE	Cash and short-term investments
ACT	Current assets
LCT	Current liabilities
INVT	Inventories
PPENT	Net property, plant and equipment
INTAN	Intangible assets
GDWL	Goodwill
EMP	Number of employees
DP	Depreciation and amortization

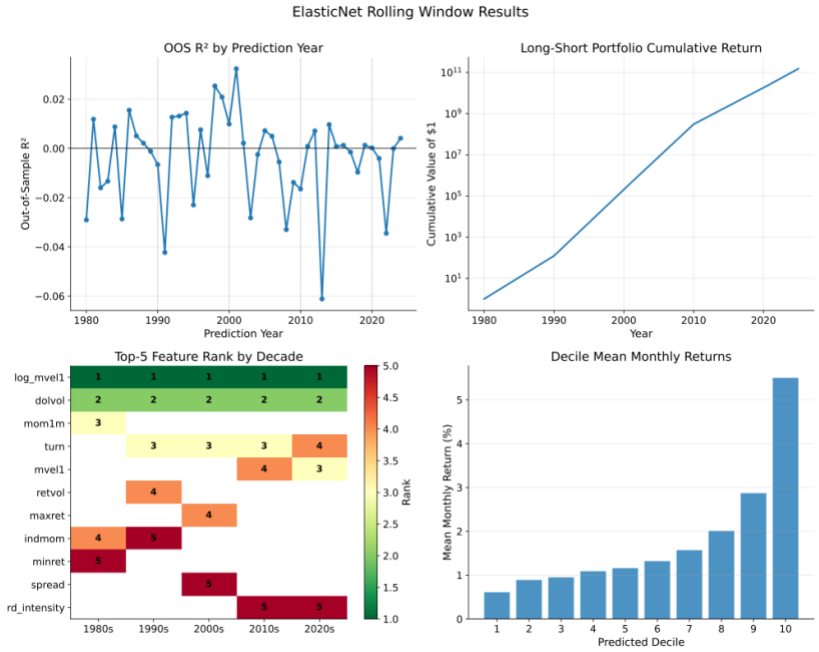
12. Appendix B

Model Diagnostics

This appendix presents full rolling window diagnostics for each model separately.

Figure 10

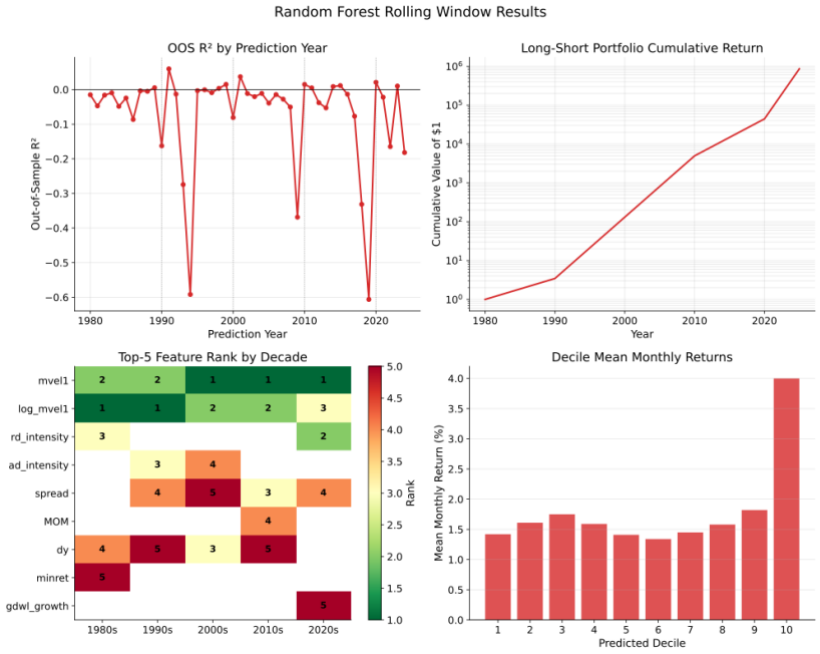
ElasticNet Rolling Window Results



Note. Each panel reports a different diagnostic for the ElasticNet model across all 45 rolling windows (1980–2024). Top-left: cumulative growth of $1 invested in the long-short portfolio (log scale). Top-right: out-of-sample R^2 per prediction year, with blue (red) bars indicating positive (negative) values and the dashed line indicating the sample mean. Bottom-left: feature importance heatmap based on mean absolute coefficient values across windows. Bottom-right: number of non-zero coefficients selected per window, with the dashed line indicating the total number of features (49).

Figure 11

Random Forest Rolling Window Results



Note. Each panel reports a different diagnostic for the Random Forest model across all 45 rolling windows (1980–2024). Top-left: cumulative growth of $1 invested in the long-short portfolio (log scale). Top-right: out-of-sample R^2 per prediction year. Bottom-left: feature importance heatmap based on permutation importance. Bottom-right: mean monthly return by predicted decile (D1 = lowest, D10 = highest predicted return).

Figure 12

Neural Network Rolling Window Results



Note. Each panel reports a different diagnostic for the Neural Network model across all 45 rolling windows (1980–2024). Top-left: cumulative growth of $1 invested in the long-short portfolio (log scale). Top-right: out-of-sample R^2 per prediction year. Bottom-left: feature importance heatmap based on permutation importance. Bottom-right: mean monthly return by predicted decile (D1 = lowest, D10 = highest predicted return).

13. Appendix C

Hyperparameter Selection and Robustness

This appendix complements the methodology and results sections by reporting the hyperparameter values selected for each decade and an additional robustness figure for the alpha-grid analysis. Together these materials make the modeling choices fully transparent and confirm that the patterns documented in the main text are not artifacts of a particular parameterization.

Table 11

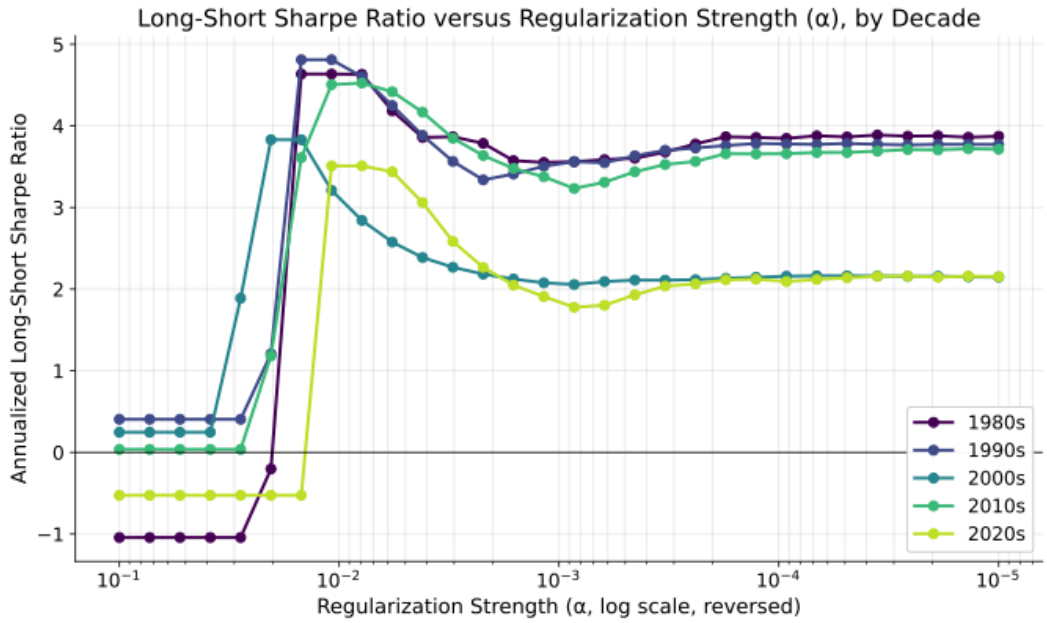
Decade Hyperparameter Selections

Decade	RF max_depth	RF min_samples_leaf	RF max_features	NN dropout_rate	NN l1_lambda
1980s	6	1000	0.5	0.20	0.0001
1990s	4	500	0.5	0.20	0.00001
2000s	6	500	0.5	0.20	0.00001
2010s	8	500	0.5	0.05	0.00001
2020s	4	500	0.5	0.20	0.00001

Note. Hyperparameters were selected separately for each decade through a single train/validation split, as described in Section 4.4. The Random Forest grid considered max_depth in {4, 6, 8, 10}, min_samples_leaf in {500, 1000, 2000}, and max_features in {sqrt, 0.5}. The Neural Network grid considered dropout_rate in {0.05, 0.20} and l1_lambda in {0.00001, 0.0001}. The combination with the highest validation R^2 in each decade was retained for all rolling windows in that decade.

Figure 13

Long-Short Sharpe Ratio versus Regularization Strength (α)



Note. Each line plots the annualized Sharpe ratio of the ElasticNet long-short decile portfolio as α varies along a logarithmic grid from 0.1 to 0.00001, computed separately for each decade. The x-axis is reversed so that regularization weakens from left to right. This figure presents the same alpha-grid result as Figure 9 with α on the x-axis instead of the number of non-zero coefficients.

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